Q1. From a morphological point of view, there are departments in the nervous system (NS):

A) central NS (CNS), vegetative NS (VNS);

B) CNS, somatic NS (SNS);

C) somatic NS, VNS;

✓ D) CNS and peripheral NS (PNS);

Q2. The organs of the central nervous system include:

A) spinal ganglia;

B) peripheral nerves;

C) intramural ganglia;

✓ D) brain.

Q3. The cytoarchitectonics of the cerebral cortex is:

A) the regular arrangement of Betz cells;

B) the regular arrangement of the nerve fibers of the cortex ;

✓ C) the regular arrangement of cortical neurocytes;

D) the natural location of the neuroglia.

Q4. The column (module) of the cerebral cortex is:

A) a structural element of the cortex;

B) functional element of the bark

✓ C) structural and functional element of the cortex;

D) part of the hemato-encephalic barrier.

Q5. The layer of Betz cells (ganglionic) is formed by neurons:

A) pear-shaped;

✓ B) pyramidal;

C) stellate;

D) basket-shaped.

Q6. The layer of the cerebral cortex containing large pyramid-shaped neurons:

A) molecular;

✓ B) ganglionic;

C) granular;

D) polymorphic.

Q7. A layer of the cerebral cortex containing cells of various shapes:

A) granular;

B) pyramidal;

C) ganglionic;

✓ D) polymorphic.

Q8. The ventricles of the brain and the central canal of the spinal cord are lined with cells:

A) astrocytes;

✓ B) ependymocytes;

C) oligodendrogliocytes;

D) microglia.

Q9. The outer layer of the cerebellar cortex is called:

A) polymphic;

✓ B) molecular;

C) pyramidal;

D) granular.

Q10. Axons of basket neurocytes of the cerebellum form synapses with cells:

A) Betsa;

✓ B) Purkinje;

C) stellate;

D) grains.

Q11. The efferent inhibitory pathways of the cerebellum are formed by cells:

A) pyramidal;

✓ B) pear-shaped;

C) basket-shaped;

D) stellate.

Q12. Climbing nerve fibers in the cerebellum end in:

✓ A) pear-shaped cells;

B) basket cells;

C) grain cells;

D) Golgi cells.

Q13. Motor neurons of the spinal cord are located in:

A) the posterior horns;

✓ B) front horns;

C) side horns;

D) rear ropes.

Q14. Through the posterior roots of the spinal cord pass:

✓ A) afferent nerve fibers;

B) efferent nerve fibers.

C) preganglionic nerve fibers;

D) postganglionic nerve fibers.

Q15. The posterior horns of the spinal cord contain:

A) motor neurons;

B) sensory neurons;

✓ C) associative neurons;

D) autonomic nuclei.

Q16. Clark's nucleus (thoracic nucleus) of the spinal cord is located in:

A) anterior horns;

B) lateral horns;

C) side ropes;

✓ D) rear horns.

Q17. The autonomic nerve centers are located in the following structure of the spinal

A) cord:

B) in the posterior horns of gray matter;

✓ C) in the lateral horns of gray matter;

D) in the anterior horns of gray matter;

E) in white matter.

Q18. The centers of the sympathetic division of the autonomic nervous system are located:

A) in the brain stem;

B) in the paravertebral ganglia;

C) in the prevertebral ganglia;

✓ D) in the lateral horns of the thoracolumbal spinal cord.

Q19. The arachnoid membrane of the spinal cord is formed by:

A) reticular tissue

✓ B) loose fibrous unformed connective tissue;

C) dense fibrous shaped tissue;

D) dense fibrous unformed tissue.

Q20. The shell adjacent to the white matter of the spinal cord is called:

A) arachnoid;

B) dura mater;

C) perineurium;

✓ D) soft brain.

Q21. Where are the bodies of sensitive neurons located?

✓ A) in the spinal nodes;

B) in the posterior horns of the spinal cord;

C) in the anterior horns of the spinal cord;

D) in the lateral horns of the spinal cord.

Q22. Where are the sensitive neurocytes innervating skeletal

A) muscles?

B) in the anterior horns of the spinal cord;

C) in the posterior horns of the spinal cord.

D) in the anterior roots of the spinal cord;

✓ E) in the spinal ganglia.

Q23. Pseudounipolar neurons are located:

A) in the cerebellum;

B) in the cerebral cortex;

C) in the vegetative ganglion;

✓ D) in the spinal ganglion.

Q24. Spinal ganglia neurocytes are surrounded by:

A) fibrous astrocytes;

B) plasma astrocytes;

✓ C) oligodendrogliocytes;

D) microglia.

Q25. Which cells of the neuroglia are surrounded by neurons of the spinal nodes?

A) astrocytes;

B) microglia;

C) ependymocytes;

✓ D) satellite oligodendrogliocytes.

Q26. The anterior horns of the spinal cord contain neurons:

A) sensitive;

✓ B) motor;

C) secretory;

D) afferent.

Q27. The anterior horns of the spinal cord contain:

✓ A) motor neurons;

B) sensory neurons;

C) associative and commissural neurons;

D) autonomic nuclei.

Q28. All of the following are the components of gray matter; except:

A) Cell bodies

✓ B) Myelinated nerve fiber

C) Neuroglia

D) Dendrites

Q29. Lateral horn is present in ____ segment of spinal cord.

A) Cervical

✓ B) Thorax

C) Lumbar

D) Sacral

Q30. Which of the following is not a layer of cerebral cortex?

A) External granular

B) External pyramidal

C) Plexiform

✓ D) Purkinje

Q31. Outer band of Baillarger is present in which layer of cerebral cortex?

A) Outer pyramidal

B) Outer granular

C) Inner pyramidal

✓ D) Inner granular

Q32. Output neuron of cerebellar cortex is:

✓ A) Purkinje

B) Stellate

C) Granule

D) Golgi Sensory organs

Q33. Name the source of the development of the anterior epithelium of the cornea of the eye:

A) mesoderm;

B) neural tube;

C) mesenchyme;

✓ D) ectoderm.

Q34. What is the source of the development of the cornea's own substance?

A) mesoderm;

B) endoderm;

✓ C) mesenchyme;

D) ectoderm.

Q35. Name the source of the development of the posterior epithelium of the cornea of the eye:

A) ectoderm;

B) neural tube;

✓ C) mesenchyme;

D) mesoderm.

Q36. Which of the listed types of epithelium belongs to the anterior epithelium of the cornea of the eye?

A) single-layer flat;

B) single-layer cubic;

C) single-layer multi-row cylindrical;

✓ D) multilayered non-corneating.

Q37. What is the receptor apparatus of the eye?

A) cornea;

B) sclera;

C) the lens;

✓ D) the retina.

Q38. What applies to the accommodation apparatus of the eye?

A) retina;

✓ B) cornea;

C) the iris;

D) the sclera.

Q39. The anterior part of the vascular membrane, which is a pigmented disk with a hole in the center(pupil) is called:

A) the cornea;

B) the anterior chamber of the eye;

✓ C) the iris;

D) the actual vascular membrane.

Q40. The ciliary muscle, ciliary processes with vessels, externally covered with the ciliary part of the retina are:

A) pupil;

B) iris;

✓ C) the ciliary body;

D) the lens.

Q41. How many neurons make up the chain of the receptor apparatus of the eye?

A) one;

B) two;

✓ C) three;

D) four.

Q42. Which of the listed cells will tilt to the photoreceptors of the retina?

A) bipolar;

B) multipolar;

C) pigmented;

✓ D) cones.

Q43. Which germ cells give rise to cone-bearing and rod-bearing cells?

A) myoblasts;

B) melanoblasts;

✓ C) neuroblasts;

D) fibroblasts.

Q44. Specify the neurites of which retinal cells form a layer of nerve fibers?

A) amacrine;

B) horizontal;

C) associative;

✓ D) ganglionic.

Q45. Which vitamin is a necessary component of the visual pigment rhodopsin?

A) vitamin B;

B) vitamin PP;

✓ C) vitamin A;

D) vitamin D.

Q46. Which layer of the retina in the central fossa (the place of the best vision) does not move apart for the course of light rays tothe layer of cones?

✓ A) ganglion;

B) external granular;

C) internal granular;

D) external mesh.

Q47. A person has impaired twilight vision ("chicken blindness"). Which cells' function is impaired?

A) cones;

✓ B) sticks;

C) amacrine;

D) horizontal.

Q48. Name the number of cells of the receptor layer in the olfactory organ:

A) one;

B) two;

✓ C) three;

D) four.

Q49. What special organoid do olfactory cells have?

✓ A) cilia;

B) flagella;

C) microvilli;

D) tonofibrils.

Q50. A person has a damaged mucous membrane covering the upper part of the middle shell of the nasal cavity.

A) The peripheral part of which analyzer is destroyed in this case?

B) taste;

✓ C) sense of smell;

D) hearing;

E) balance.

Q51. What type of receptor is the taste organ?

A) baroreceptor;

✓ B) chemoreceptor;

C) thermoreceptor;

D) mechanoreceptor.

Q52. As a result of accidental use of acetic acid, the taste buds of the anterior

A) part of the tongue atrophied in the patient. What taste irritations are lost in this case?

✓ B) sweet;

C) salty;

D) bitter;

E) sour.

Q53. A person has affected taste buds at the root of the tongue, what taste

A) sensations are preserved (violated?)?

B) sweet;

✓ C) bitter;

D) sour;

E) salty.

Q54. In which part of the ear is the cortical organ?

A) outer ear;

B) middle ear;

✓ C) a snail;

D) a system of semicircular channels.

Q55. What type of epithelium lines the vestibular part of the membranous labyrinth?

A) single-layer multi-row prismatic;

✓ B) single-layer single-row flat;

C) multilayer cubic;

Q56. In what place of the transverse curl of the cochlea lies the cortical organ?

A) the drum ladder;

✓ B) the membranous labyrinth;

C) vestibular staircase;

D) spiral node.

Q57. Which cells of the cortical organ are secondary receptor cells?

✓ A) internal hairline;

B) columnar cells;

C) supporting cells;

D) vascular strip cells.

Q58. What is the organelle of special significance in the receptor cells of the cortical organ?

A) kinocilia;

B) neurofibrils;

✓ C) microvilli;

D) flagella.

Q59. Which cells of the cortical organ form a tunnel?

A) external hair cells;

B) external supporting;

✓ C) column cells;

D) internal hair cells.

Q60. Which cells of the spiral organ do not lie on the basement membrane?

✓ A) external and internal hair cells;

B) column cells;

C) external supportive;

D) internal supportive.

Q61. Which cells of the spiral organ transmit excitation to the spiral node?

A) cells-pillars;

B) external supporting;

C) internal supportive;

✓ D) external and internal hair.

Q62. In which part of the ear is the organ of balance?

A) the outer ear;

B) the middle ear;

C) the cochlea;

✓ D) the vestibular labyrinth.

Q63. Outer epithelial layer of tympanic membrane is lined by this epithelium.

A) Transitional

B) Pseudostratified

C) Stratified squamous keratinized

✓ D) Stratified squamous nonkeratinized

Q64. All are parts of membranous labyrinth, except:

✓ A) Vestibule

B) Saccule

C) Semicircular duct

D) Cochlear duct

Q65. Which structure is transparent?

A) Choroid

B) Ciliary body

C) Iris

✓ D) Cornea

Q66. What type of neurons is present in the retina?

A) Unipolar

B) Pseudounipolar

✓ C) Bipolar

D) Multipolar

Q67. Which layer of the cornea is acellular?

A) Epithelium

B) Endothelium

✓ C) Descemet’s membrane

D) Substantia propria

Q68. Which of the following is the receptor for color?

A) Rods

✓ B) Cones

C) Bipolar cells

D) Ganglion cells

Q69. All of the following are the functions of pigment layer, except:

A) It absorbs and prevents back reflection of light

✓ B) Convert light to electrical energy

C) Provide mechanical support to them

D) They have a phagocytic role

Q70. Cell that form the internal limiting membrane is:

A) Ganglion

✓ B) Müller

C) Bipolar

D) Amacrine

Q71. Keratitis is a inflammation of:

A) Sclera

B) Retina

✓ C) Cornea

D) Choroid CARDIOVASCULAR SYSTEM. LYMPHOID ORGANS.

Q72. List what the cardiovascular system consists of?

A) arteries, veins, venules, capillaries, lymphatic capillaries

✓ B) heart, arteries, veins, capillaries, venules, arteriol-venular anastomoses, lymphatic capillaries, vessels, ducts

C) heart, lymph nodes, veins, capillaries, venules, lymphatic capillaries

D) heart, arteries, veins, lymphatic capillaries, lymphatic vessels and ducts

E) arteries, capillaries, venules, veins, lymphatic vessels and ducts, microcirculatory bed

Q73. Choose, the sources of heart development are ...

A) the visceral leaf of the mesoderm

✓ B) mesenchyme and visceral leaf of mesoderm

C) mesenchyme and parietal leaf of mesoderm

D) endoderm of the primary intestine

E) nervous crest

Q74. Specify the time of laying the heart during embryogenesis ...

A) at 4 weeks

B) at week 5

✓ C) at 3 weeks

D) at week 6

E) at 2 weeks

Q75. Indicate whether the blood vessels in the human embryo develop from ...

A) the mesenchyme of the yolk sac

B) mesenchyma of the trunk

✓ C) the mesenchyme of the yolk sac and the mesenchyme of the trunk

D) endoderms of the primary intestine

E) endoderms

Q76. List what factors determine the structure of blood vessels?

A) the activity of biologically active substances

✓ B) blood pressure, blood flow rate

C) blood flow velocity, gravity

D) gravity, circulating immunoglobulins in the blood

E) the influence of the central organs of hematopoiesis

Q77. Name what tissues form the artery wall?

✓ **A) epithelial, smooth muscle, loose connective tissue**

B) epithelial, striated muscle tissue, loose connective tissue

C) epithelial, smooth muscle tissue, dense decorated connective tissue

D) epithelial, smooth muscle tissue

E) epithelial, smooth muscle tissue, reticular

Q78. Specify what vessels are characterized by the presence of internal and external elastic membranes?

A) muscle type vein

B) arteriole

✓ **C) muscle-type artery**

D) lymphatic vessel

E) muscle venule

Q79. Histological specimen presents a vessel the wall of which consists of endothelium, basal membrane and loose connective tissue. What type of vessel is it?

✓ **A) Hemocapillary**

B) Lymphocapillary

C) Artery

D) Vein of muscular type

E) Vein of non-muscular type

Q80. An electronic microphotograph shows a macrophagic cell with erythrocytes at different stages of differentiation located along its processes. This is the cell of the following organ:

✓ **A) Spleen**

B) Thymus

C) Red bone marrow

D) Tonsil

E) Lymph node

Q81. Live vaccine is injected into the human body. Increasing activity of what cells of connective tissue can be expected?

✓ **A) Plasmocytes and lymphocytes**

B) Pigmentocytes and pericytes

C) Fibroblasts and labrocytes

D) Adipocytes and adventitious cells

E) Macrophages and fibroblasts

Q82. In the microspecimen of red bone marrow there were revealed multiple capillares through the walls of which mature blood cells penetrated. What type of capillares is it?

A) Somatical

B) Fenestrational

✓ C) Sinusoidal

D) Visceral

E) Lymphatic

Q83. Blood sampling for bulk analysis is recommended to be performed on an empty stomack and in the morning. What changes in blood composition can occur if to perform blood sampling after food intake?

✓ A) Increased contents of leukocytes

B) Increased plasma proteins

C) Reduced contents of erythrocytes

D) Reduced contents of thrombocytes

E) Increased contents of erythrocytes

Q84. Low level of albumins and fibrinogen was detected in the patients blood. Decreased activity of what organelle of the liver hepatocytes can cause it?

A) Lysosomes

✓ B) Granular endoplasmatic reticulum

C) Mitochondrions

D) Agranular endoplasmatic reticulum

E) Golgi complex

Q85. In course of an experiment a big number of stem cells of red bone marrow was in some way destructed. Regeneration of which cell populations in the loose connective tissue will be inhibited?

A) Of fibroblasts

B) Of lipocytes

C) Of pericytes

✓ D) Of macrophags

E) Of pigment cells

Q86. In the histological preparation of the vessels, the internal and external elastic membranes are well expressed and there are many myocytes in the middle shell. What vessel are we talking about?

A) Extraaortic lymphatic system;

B) A vein with strong muscle development;

C) Elastic type artery;

D) Mixed type artery;

✓ E) Muscle-type artery

Q87. Large-caliber arteries during systole stretch and return to their original state during diastole, ensuring the stability of blood flow. What are the elements of the vessel wall that can explain this?

A) A large number of fibroblasts;

✓ B) Elastic fibers;

C) Reticular fibers;

D) Muscle fibers;

E) Collagen fibers.

Q88. If there are terminal elastic membranes in the middle shell of the vessel so this is...

A) arteriole

B) venula

C) mixed type artery

✓ D) muscle-type artery

E) elastic type artery

Q89. The inner shell of the vessel (intima) is lined with epithelium from the inside. Name it?

A) Mesothelium;

B) Transitional epithelium;

C) Multi-row epithelium;

✓ D) Endothelium;

E) The epidermis.

Q90. The heart wall is represented in the micropreparation. In one of the shells there are contractile, conductive and secretory myocytes, endomysium with blood vessels. Specify which shell of the heart these structures belong to?

✓ A) atrial myocardium;

B) ventricular endocardium;

C) adventitial;

D) pericardium;

E) epicardium of the heart

Q91. Choose which of the vessels belongs to the microcirculatory bed…

A) arteries

B) the heart

C) lymphatic vessels

✓ D) arteriole

E) muscle-free vein

Q92. Patient A. 40 years old suffered a myocardial infarction. Due to what morphological components did the regeneration of the heart wall occur?

A) Proliferation of contractile and conductive cardiomyocytes;

✓ B) Proliferation of connective tissue cells;

C) Proliferation of contractile cardiomyocytes;

D) Intracellular regeneration of contractile cardiomyocytes;

E) Proliferation of conductive cardiomyocytes

Q93. Choose which type this capillary belongs to: endotheliocytes are large, the basement membrane and pericytes are absent, there are sling filaments?

A) postcapillary venule

B) sinusoidal capillary

✓ C) lymphatic capillary

D) fenestrated capillary

E) somatic capillary

Q94. Specify which tissues, including the heart wall is formed?

A) smooth muscle tissue

✓ B) cross-striated cardiac muscle tissue

C) cross-striated skeletal muscle tissue

D) cross-striated cardiac and smooth muscle tissues

E) transversely striated cardiac and transversely striated skeletal muscle tissue

Q95. The capillary electronogram clearly defines the fenestras in the endothelium and the time in the basement membrane. What is the type of capillary?

A) sinusoidal;

✓ B) visceral;

C) shunt;

D) atypical;

E) somatic.

Q96. Indicate in the cleavage of the basement membrane, the capillary walls are located:

A) myocytes

✓ B) pericytes

C) fibroblasts

D) adventitious cells

E) lipocytes

Q97. List, the hemocapillary wall contains:

✓ A) endotheliocytes on the basement membrane, pericytes, adventitial cells

B) smooth myocytes

C) fibroblasts

D) internal elastic membrane

Q98. Choose, the tissue composition of the artery wall of various types is determined by:

✓ A) blood pressure, the direction of blood flow

B) the presence of tissues

C) the speed of blood flow

D) the number of shells

Q99. Choose, age-related changes in the artery wall consist in:

✓ A) its compaction

B) accumulation of sulfated glycosaminoglycans

C) thickening of collagen fibers

D) thickening of elastic fibers and membranes

Q100. Specify what is characteristic of arteriol-venular anastomosis with epithelioid cells?

✓ A) longitudinal bundles of smooth myocytes in the inner shell

B) circular bundles of smooth myocytes in the outer shell

C) circular bundles of smooth myocytes in the inner shell

D) longitudinal beams in the middle shell

Q101. Choose where smooth myocytes are located in large lymphatic vessels:

A) in the inner shell

B) in the middle shell

C) in the outer shell

✓ D) in all shells

E) missing

Q102. Choose where there are vessels of vessels?

A) arteries

B) veins

C) lymphatic vessels

D) in veins and lymphatic vessels

✓ E) in all vessels

Q103. Specify what is represented by the contractile apparatus of the smooth muscle cell as part of the inferior vena cava?

A) myofibrils

✓ B) thick, thin and intermediate myofilaments

C) thick myofilaments

D) thin myofilaments

E) intermediate myofilaments

Q104. The cortical substance of the thymus is formed by the following cells:

✓ A) T-blasts and T-lymphocytes

B) plasmocytes

C) by NK cells

D) macrophages

Q105. Highlight what is the source of thymus development?

A) the epithelium of the pharyngeal intestine between the I and II pairs of gill pockets

✓ B) epithelium of the pharyngeal intestine in the area of III and IV pairs of gill pockets

C) ectodermal epithelium of the oral fossa of the embryo

D) endoderm of the primary intestine

E) coelomic epithelium of primary kidneys

Q106. Select what tissue forms the thymus stroma?

A) lymphoid

✓ B) epithelioretic

C) myeloid

D) mucosa

E) pigmented

Q107. Specify where the epithelial bodies of Ghassal are located?

A) in the cortical substance of the lymph node

✓ B) the medulla of the thymus lobule

C) in the lymphatic follicles

D) red bone marrow

E) in the spleen

Q108. Choose which hormone is synthesized in the thymus?

A) thyroxine

✓ B) thymosin

C) testosterone

D) adrenaline

E) oxytocin

Q109. Specify what causes the removal of the thymus in newborn animals?

A) increased proliferation of lymphocytes in all lymphoid nodules of hematopoietic organs

✓ B) a sharp inhibition of lymphocyte proliferation in all lymphoid nodules of hematopoietic organs

C) strengthening the synthesis of pituitary hormones

D) acceleration of puberty

E) weakening of the activity of the red bone marrow

Q110. List, the hematothymus barrier includes ...

A) epithelioretic cells - macrophages - lymphocytes

✓ B) capillary endothelium - capillary basement membrane - pericapillary space - epithelioretic cell basement membrane - epithelioretic cell cytoplasm

C) lymphoblast - perivascular cells - basement membrane - lymphocyte

D) secretory cells - perivascular cells - basal membrane of the capillary endothelium - lymphocytes

E) capillary endothelium - pericapillary space - epithelioretic cells

Q111. Alien graft rejection does not occur in mice with removed thymus. Name it, this is due to the absence of ...

A) B-lymphocytes

B) macrophages

✓ C) T-killers

D) monocytes

E) plasmocytes

Q112. An electron micrograph shows a process-shaped cell containing differentiating lymphocytes in deep invaginations of the plasmolemma. Specify which organ is characterized by such an ultrastructure?

✓ A) thymus;

B) spleen;

C) liver;

D) amygdala;

E) red bone marrow

Q113. Choose what function does the lymph node perform?

A) performs myelopoiesis

✓ B) cleanses the lymph from foreign particles and enriches it with lymphocytes

C) destroys old red blood cells

D) synthesizes thymosin

E) regulates blood clotting

Q114. Specify the source of the development of lymph nodes ...

A) visceral splanchnotoma leaf

B) mesenchyme of the primary kidney

✓ C) mesenchyma around lymphatic vessels

D) epithelium of the primary intestine

E) the parietal leaf of the splanchnotome

Q115. Identify which cells are part of the lymphatic follicle of the lymph node?

✓ A) lymphoblasts, B-lymphocytes, macrophages

B) endotheliocytes, pericytes

C) fibroblasts, fat cells

D) tanicytes, ependymocytes

E) osteocytes, process cells

Q116. The histopreparation presents an organ in which lymphocytes form three types of lymphoid structures: lymph nodes, cerebral cords and sinuses. Which body is represented?

A) thymus;

B) spleen;

✓ C) lymph node;

D) amygdala;

E) red bone marrow

Q117. The preparation contains an organ in the reticular stroma of which mature shaped blood elements are located and lymphoid formations are visible. Determine which organ is represented on the drug?

A) a) thymus;

B) b) spleen;

C) c) lymph node;

D) d) amygdala;

✓ E) e) red bone marrow.

Q118. The micropreparation shows an organ of lobular structure, the stroma of which consists of process-shaped epithelial cells. Which body is represented?

✓ A) a) thymus;

B) b) spleen;

C) c) lymph node;

D) d) amygdala;

E) e) red bone marrow

Q119. In the punctate of the myeloid tissue of a 6-year-old child, cells are found in which pyknosis and removal of the nucleus occur during differentiation. What is the type of hematopoiesis for which these morphological changes are characteristic?

A) a) thrombocytopoiesis;

B) b) lymphocytopoiesis;

C) c) monocytopoiesis;

✓ D) d) erythrocytopoiesis;

E) e) granulocytopoiesis

Q120. Foci of increased plasmocytogenesis were found in the biopsy of the lymph node in the cerebral cords. Specify the antigen-dependent stimulation of which immunocompetent cells caused their formation?

A) a) interdigitating cells;

✓ B) b) B-lymphocytes;

C) c) macrophages;

D) d) T-lymphocytes

E) d) dendritic cells.

Q121. Morphological studies of the spleen revealed activation of immune reactions in the body. In which structures of this organ does antigen-dependent proliferation of T-lymphocytes begin?

A) the mantle zone of the white pulp;

✓ B) the periarterial zone of the white pulp;

C) the central zone of the white pulp;

D) the marginal zone of the white pulp;

E) red pulp.

Q122. In a histological preparation, a hematopoietic organ consisting of particles of various shapes is examined. In each lobule there is a cortical and cerebral substance. Which organ do these signs belong to?

A) a) thymus;

B) b) spleen;

✓ C) c) lymph node;

D) d) amygdala;

E) e) vermiform process

Q123. Following are the layers of blood vessels, except:

A) Tunica adventitia

B) Tunica media

✓ C) Tunica albuginea

D) Tunica intima

Q124. Pericytes are seen in:

A) Elastic artery

B) Muscular artery

C) Large-sized vein

✓ D) Postcapillary venules

Q125. Vessels responsible for peripheral resistance are:

A) Elastic artery

✓ B) Arterioles

C) Large-sized vein

D) Venules

Q126. Deep vein thrombosis mainly involves:

A) Elastic artery

✓ B) Medium-sized vein

C) Large-sized vein

D) Postcapillary venules THE ENDOCRINE SYSTEM

Q127. Specify the sources of thyroid development?

✓ A) the ventral dorsum of the pharyngeal intestine between the I and II pairs of gill pockets

B) the dorsal wall of the pharyngeal intestine between the I and II pairs of gill pockets

C) epithelium III and IV pairs of gill pockets

D) thickening of the coelomic epithelium of the mesentery root

E) ectodermal epithelium and ganglion plate

Q128. Specify the sources of development of the adrenal glands ...

A) mesonephral duct and nephrogenic tissue

✓ B) thickening of the coelomic epithelium and neuroblasts of the rudiments of sympathetic ganglia

C) mesenchyme in the thickness of the dorsal mesentery

D) visceral splanchnotome leaf

E) nephrogonotomy

Q129. Specify the sources of development of the epiphysis ...

A) ganglion plate

B) the dorsal wall of the medulla oblongata

✓ C) the dorsal wall of the intermediate brain

D) distal end of the funnel of the third ventricle

E) the midbrain

Q130. Choose which organ contains neurosecretory cells whose processes have extensions containing secretory granules and form synapses with vessels of the neurohypophysis?

A) the pituitary gland

B) epiphysis

✓ C) hypothalamus

D) cerebellum

E) the medulla oblongata

Q131. Tell us what function does the pituitary gland perform?

A) synthesizes adrenaline

✓ B) regulates the activity of thyrocytes

C) is the central organ of immunogenesis

D) participates in hematopoiesis and immunogenesis

E) is a peripheral organ of immunogenesis

Q132. Name which cells synthesize follicle-stimulating hormone?

A) thyrotropocytes

B) pituities

C) somatotropocytes

✓ D) gonadotropocytes

E) corticotropocytes

Q133. Specify which cells synthesize thyroid-stimulating hormone?

A) chromophobic cells of the adenohypophysis

✓ B) thyrotropocytes

C) follicular-stellate cells

D) pituicities

E) neurosecretory cells of the supraoptic nucleus

Q134. Specify which cells synthesize adrenocorticotropic hormone?

A) endocrinocytes of the glomerular zone of the adrenal cortex

✓ B) corticotropocytes

C) somatotropocytes

D) pinealocytes

E) calcitoninocytes

Q135. Choose which cells regenerate the adenohypophysis?

A) gonadotropocytes

✓ B) chromophobic cells of the adenohypophysis

C) follicular-stellate cells

D) thyrotropocytes

E) pinealocytes

Q136. Highlight, the specificity of the action of hormones is determined by ...

A) the half-life of the hormone in the vascular bed

✓ B) the presence of hormone receptors in cells

C) the nature of the endothelium of blood capillaries in the target tissue

D) the concentration of the hormone in the blood

E) the rhythm of hormone secretion by the endocrinocyte

Q137. Choose, thyrocytes secrete is...

A) thyroid-stimulating hormone

✓ B) thyroxine

C) parathyrin

D) calcitonin

E) serotonin

Q138. Name, calcitoninocytes of the thyroid gland secrete ...

A) parathyrin

✓ B) calcitonin

C) thyroid-stimulating hormone

D) somatotropic hormone

E) prolactin

Q139. Specify what function does the adrenal cortex perform?

✓ A) synthesizes corticosteroids

B) synthesize catecholamines

C) carries out the absorption of vitamins

D) performs antigen-independent differentiation of T-lymphocytes

E) synthesizes thyroid-stimulating hormone

Q140. Specify which large-cell neurosecretory nuclei of the hypothalamus include:

A) ventromedial

B) arcuate

✓ C) supraoptic

D) dorsomedial

E) periventricular

Q141. Choose, the specificity of the action of hormones depends on:

A) chemical composition

B) blood concentrations

C) binding to the carrier protein

D) metabolic rates in tissues

✓ E) the presence of receptors on target cells

Q142. A parenchymal organ is determined on a histological preparation. The structural and functional unit of which is the follicle. The follicle wall is formed by cubical cells, the follicle cavity is filled with a colloid. Choose which organ is represented in the preparation?

A) pituitary gland;

B) ovary;

✓ C) thyroid gland;

D) adrenal glands;

E) testes

Q143. Indicate, if there is a lack of iodine in the body, the formation of hormones is disrupted:

A) the epiphysis

B) adenohypophysis

C) the adrenal glands

✓ D) thyroid gland

E) parathyroid glands

Q144. From the ectodermal epithelium of the lining of the upper part of the oral fossa of the human embryo, a Ratke pocket is formed, which is directed to the base of the future brain. Specify what develops from this embryonic germ?

A) Anterior hypothalamus;

✓ B) Adenohypophysis;

C) Medial elevation;

D) Neurohypophysis;

E) Pituitary pedicle

Q145. Choose, the thyroid gland is formed from:

A) mesenchyma

B) neuroblasts of nerve crests

✓ C) pharyngeal epithelium

D) skin ectoderm

Q146. Indicate whether the microcirculatory bed of the endocrine glands is characterized by the presence of:

A) sinusoidal capillaries

✓ B) fenestrated endothelium in capillaries

C) developed pericapillary spaces

D) precapillary sphincters

Q147. Finish the definition. Herring's accumulative corpuscles in the neurohypophysis are:

A) the endings of gliocyte processes on the basal membranes of blood vessels

B) clusters of pituitary

C) dilated and overflowing with blood hemocapillaries

✓ D) axon terminals with neural secret

Q148. What function does the adrenal medulla perform?

A) carries out the absorption of vitamins

B) synthesizes corticosteroids

✓ C) synthesizes catecholamines

D) performs antigen-independent differentiation of T-lymphocytes

E) synthesizes thyroid-stimulating hormone

Q149. The patient was given high doses of hydrocortisone for a long time, as a result of which atrophy of one of the zones of the adrenal cortex occurred. Specify which zone is this?

✓ A) fasicular;

B) reticular;

C) glomerular;

D) glomerular and reticular;

Q150. When examining one of the adrenal glands removed during surgery, large cells were found that were impregnated with a solution of potassium bicarbonate. Choose which hormone these cells synthesize?

A) Cholecystokinin;

B) Aldosterone;

C) Thyroxine;

✓ D) Adrenaline;

E) thymosin.

Q151. In the wall of the follicles and in the interfollicular layers of connective tissue on the territory of the thyroid gland, large endocrinocytes are placed, the secretory granules of which are osmium and argyrophilic. Name these cells?

A) Pinealocytes;

B) Pituicites;

✓ C) Calcitoninocytes;

D) Thyrocytes;

E) Parathyrocytes.

Q152. The parenchyma of the adenohypophysis is represented by trabeculae formed by glandular cells. Among the adenocytes there are cells with granules that are stained with basic dyes and contain glycoproteins. What are these cells?

A) Somatotropocytes;

B) Mammotropocytes;

C) Chromophobic;

✓ D) Gonadotropocytes, thyrotropocytes;

E) Melanotropocytes.

Q153. In the endocrinology department, the patient was diagnosed with acromegaly. Choose the hyperfunction of which pituitary cells is caused by this disease?

A) Thyrotropocytes;

✓ B) Somatotropocytes;

C) Chromophobic;

D) Gonadotropocytes;

E) Mammothropocytes.

Q154. A 50-year-old male farm worker has been brought to the emergency room. He was found confused in the orchard and since then has remained unconscious. His heart rate is 45 and his blood pressure is 80/40 mm Hg. He is sweating and salivating profusely. Which of the following should be prescribed?

A) Norepinephrine,

B) Physostigmine,

C) Pentamine

✓ D) Atropine

E) Proserine

Q155. A patient has been given high doses of hydrocortisone for a long time. This caused atrophy of one of the adrenal cortex zones. Which zone is it?

✓ A) zona fasciculata

B) zona reticularis

C) zona glomerulosa and zona reticularis

D) zona glomerulosa

Q156. Select the hormones produced by the oxyphilic cells of the adenohypophysis

A) oxytocin, vasopressin;

✓ B) somatotropin, prolactin

C) thyrotropic

D) gonadotropic

E) corticotropic

Q157. Identify what pinealocytes synthesize

A) thyrotropin

B) calcitonin

C) oxytocin

✓ D) melatonin, serotonin

E) prolactin

Q158. State what determines the specificity of hormone action

A) half-life of the hormone in the vascular system

✓ B) presence of hormone receptors on target cells

C) nature of endothelium of blood capillaries in target tissue

D) hormone concentration in blood

E) rhythm of hormone secretion by endocrinocytes

Q159. A patient has sharply increased daily urine output. What hypothalamic hormone secretion deficiency could explain this phenomenon?

A) oxytocin

B) melatonin

✓ C) vasopressin

D) serotonin

E) prolactin

Q160. An animal has had its thyroid gland removed. Determine the hypertrophy of which pituitary gland cells will be found in the animal?

✓ A) basophilic endocrinocytes

B) chromophilic cells

C) oxyphilic endocrinocytes

D) pinealocytes

E) follicular cells

Q161. What is the function of the adrenal cortex?

✓ A) synthesizes corticosteroids

B) synthesizes catecholamines

C) performs vitamin absorption

D) performs antigen-independent differentiation of T-lymphocytes

E) synthesizes thyroid hormone

Q162. Determine what makes up the hypothalamic-neurohypophyseal system

A) hypothalamic arcuate complex and median eminence

B) adenohypophysis and hypothalamus

C) hypothalamus and thyroid gland

✓ D) supraoptic and paraventricular nuclei and posterior pituitary lobe

E) supraoptic and paraventricular nuclei and medial elevation

Q163. Which of the following is not a characteristic of the endocrine system?

A) Products secreted into blood

✓ B) Glands with ducts

C) Secretes hormones

✓ B) Nonlocalized response

Q164. Calcitonin is a hormone of which of following?

A) Adrenal cortex

✓ B) Thyroid gland

C) Pituitary gland

D) Thymus gland

Q165. Calcium level in the blood is regulated by the:

A) Thyroid

B) Parathyroid

C) Posterior pituitary

D) Adrenal medulla

✓ E) a and b

Q166. Which gland is called the "master gland"?

A) Adrenal medulla

B) Adrenal cortex

C) Thyroid

✓ D) Pituitary DIGESTIVE SYSTEM

Q167. Crypts are detected in the histological preparation of the palatine tonsil, the epithelium of which is infiltrated by leukocytes. Specify which epithelium is part of this organ?

✓ A) stratified squamous non-keratinizing;

B) stratified cubic;

C) stratified ciliated;

D) stratified squamous keratinizing;

E) simple prismatic.

Q168. A patient visited a dentist with complaints of redness and edema of his mouth mucous membrane in a month after dental prosthesis. The patient was diagnosed with allergic stomatitis. What type of allergic reaction by Gell and Cumbs underlies this disease?

A) Cytotoxic

B) Anaphylactic,

C) Stimulating

✓ D) Delayed type hypersensitivity

E) Immunocomplex.

Q169. When the pH level of the stomach lumen decreases to less than 3, the antrum of the stomach releases peptide that acts in paracrine fashion to inhibit gastrin release. This peptide is:

A) Gastrin-releasing peptide (GRP)

B) Acetylcholine

C) GIF

✓ D) Somatostatin

E) Vasoactive intestinal peptide (VIP)

Q170. In the histological preparation of the glandular organ, only the serous terminal sections are determined. In the interlobular connective tissue, ducts lined with a two-layer or multi-layer epithelium are visible. Identify this organ?

✓ A) parotid gland;

B) pancreas;

C) liver;

D) sublingual salivary gland;

E) submandibular salivary gland

Q171. Specify what is secreted by the main exocrinocytes of the stomach?

A) hydrochloric acid

✓ B) pepsinogen

C) serotonin

D) mucus

E) somatostatin

Q172. Choose which cells are missing in the glands of the stomach?

A) the main

B) lining

C) mucous membranes

D) G-cells

✓ E) A-cells

Q173. What cells are missing in intestinal crypts?

A) columnar epithelial cells

B) undifferentiated epithelial cells

C) goblet cells

D) Paneta acidophilic cells

✓ E) parietal cells

Q174. In inflammatory diseases of the stomach, the integumentary epithelium of the mucous membrane is damaged. Choose which epithelium suffers at the same time?

✓ A) simple columnar glandular;

B) simple cubic;

C) stratified cubic;

D) simple flat;

E) simple cubic microvilli.

Q175. In diseases of the mucous membrane of the small intestine, the absorption function suffers. Name which epithelium is responsible for this function?

✓ A) simple columnar glandular with a brush-border;

B) simple cubic;

C) stratified cubic;

D) simple flat;

E) simple cubic microvilli

Q176. When examining a patient with a small intestine disease, violations of the processes of parietal and membrane digestion were revealed. With the violation of the function of which cells is this associated?

A) goblet-shaped;

✓ B) columnar epithelium with a brush-border;

C) endocrinocytes;

D) Paneta cells;

E) columnar without a border

Q177. During the examination of the patient, an anomaly of liver development was revealed. Please indicate which embryonic source has been damaged?

A) endoderm of the posterior intestine;

B) endoderm of the middle part of the primary intestine;

✓ C) endoderm of the anterior intestine;

D) endoderm of the posterior wall of the trunk intestine;

E) mesonephral strait

Q178. Select, Paneta cells…

✓ A) lysozyme is isolated

B) cholecystokinin is isolated

C) amylase is isolated

D) lipase is isolated

E) secrete serotonin

Q179. Name, perisinusoidal lipocytes are also called …

A) Kupfer cells

B) pit cells

✓ C) Ito cells

D) Golgi cells

E) Merkel cells

Q180. Specify whether the composition of pancreatic juice includes…

A) glucagon

B) insulin

✓ C) trypsinogen

D) somatostatin

E) pancreatic polypeptide

Q181. Determine if the pitelium in the middle part of the esophagus is:

A) single-layer flat

✓ B) stratified squamous non-keratinizing epithelium

C) keratinizing

D) multi-row

E) edged

Q182. What is characteristic of the structure of the liver?

✓ A) blood from the sinus capillaries enters the interlobular vein

B) blood flows from the liver through the portal vein

C) the hepatic veins enter through the liver gate

D) hepatocytes have two free surfaces

E) hepatocytes lie on the basement membrane

Q183. During histological examination of the stomach it was found out that glands contain very small amount of pariental cells or they are totally absent. Mucose membrane of what part of the stomach was studied

A) Cardiac part

B) Fundus of stomach

✓ C) Pyloric part

D) Body of stomach

Q184. A 2-year-old child has got intestinal dysbacteriosis, which results in hemorrhagic syndrome. What is the most likely cause of hemorrhage of the child?

A) PP hypovitaminosis

B) Activation of tissue thromboplastin

✓ C) Vitamin K insufficiency

D) Fibrinogen deficiency

E) Hypocalcemia

Q185. Some diseases of the small intestine are associated with impaired function of exocrinocytes with acidophilic granules (Paneta cells). Specify where these cells are located?

A) on the lateral surfaces of intestinal villi;

B) in the upper part of the intestinal crypts;

C) at the junction of the villi in the crypt;

D) on the apical part of the intestinal villi;

✓ E) at the bottom of intestinal crypts.

Q186. In some diseases of the colon, the quantitative ratios between the epithelial cells of the mucosa change. What are the types of cells that predominate in the epithelium of the crypts of the colon normally?

✓ A) goblet cells;

B) endocrinocytes;

C) poorly differentiated cells;

D) cells with acidophilic granules;

E) columnar villous epithelial cells.

Q187. Rectoromanoscopy revealed a tumor emanating from the mucosa of the intermediate rectum. Choose from which epithelium this tumor was formed?

A) simple columnar epithelium with a border;

B) simple cuboidal epithelium;

✓ C) stratified squamous non-keratinizing

D) simple squamous epithelium;

E) simple cuboidal ciliated epithelium.

Q188. During the examination of the patient, an anomaly of liver development was revealed. Please indicate which embryonic source has been damaged?

A) endoderm of the posterior intestine;

B) endoderm of the middle part of the primary intestine;

✓ C) endoderm of the anterior intestine;

D) endoderm of the posterior wall of the trunk intestine;

E) mesonephral strait.

Q189. With the proliferation of connective tissue in the liver parenchyma (fibrosis) due to chronic diseases, there is a violation of blood circulation in the classical lobules. Determine which direction of blood flow in such lobules?

A) around the lobule;

B) from the center to the periphery;

✓ C) from the periphery to the center;

D) from the top to the base;

E) from the base to the top.

Q190. In people who are prone to excessive consumption of sweets, certain cells of the pancreas are constantly in a state of tension. Specify which ones?

A) D-cells;

B) A-cells;

✓ C) B-cells;

D) Acinose-insular;

E) PP-cells.

Q191. In the histopreparation of the small intestine, villi covered with tissue consisting only of cells forming a layer that is located on the basement membrane are determined. The tissue does not contain blood vessels. Choose which tissue covers the surface of the villi?

✓ A) epithelial tissue;

B) dense unformed connective tissue;

C) reticular tissue;

D) smooth muscle tissue;

E) loose fibrous connective tissue.

Q192. The histological preparation presents an organ of the digestive tract, the wall of which consists of 4 membranes: mucous, submucosal, muscular and serous. The mucous membrane has folds and pits. Determine which organ has this relief?

✓ A) stomach;

B) esophagus;

C) small intestine;

D) vermiform process;

E) duodenum.

Q193. The histopreparation presents an organ, in its own plate of the mucous membrane of which there are simple tubular glands, consisting mainly of the main and parietal, as well as mucous, cervical endocrine cells. Specify the type of glands?

A) cardiac glands of the esophagus;

✓ B) the fundus glands;

C) the cardiac glands of the stomach;

D) pyloric glands of the stomach;

E) esophageal glands.

Q194. The histopreparation contains iron. Acinuses are determined in the lobules, the secretory cells of which have two zones: basal-homogeneous basophilic and apical-zymogenic oxyphilic. Choose which organ has these key morphological features?

A) submandibular salivary gland;

B) sublingual salivary gland;

C) liver;

D) parotid salivary gland;

✓ E) the pancreas.

Q195. The patient was admitted to a therapeutic clinic. A decrease in the acidity of gastric juice has been established in the laboratory. Specify which cells of the gastric glands caused this condition?

✓ A) Parietal cells;

B) Cervical cells;

C) Mucous membranes;

D) Main cells;

E) Endocrine.

Q196. Morphological analysis of biopsy material of the esophageal mucosa taken from the patient revealed the process of keratinization of the epithelium. Highlight which of the following types of epithelium covers the mucous membrane of this organ normally?

A) Pseudostratified ciliated;

B) Simple squamous ngle-layer flat;

✓ C) Stratified squamous non-keratinizing

D) Simple columnar;

E) Stratified squamous keratinizing.

Q197. In a cancer patient after radiation therapy, morphological examination revealed a significant violation of the regeneration process of the epithelial layer of the mucous membrane of the small intestine. Determine which epithelial cells are damaged?

A) goblet-shaped;

B) columnar with a border;

C) endocrinocytes;

D) Paneta cells;

✓ E) columnar without a brush-border.

Q198. An infectionist doctor revealed in a syndrome of acute enterocolitis with a violation of the process of digestion and absorption of cleavage products in a patient. Specify, when the intestinal epithelial cells are damaged, such violations are observed?

A) goblet-shaped;

✓ B) columnar with a border;

C) endocrinocytes;

D) Paneta cells;

E) columnar without a brush-border.

Q199. On a histological preparation, the submucosal base of the small intestine is filled with the final secretory sections of the protein glands. Which part of the intestine is represented on the drug?

A) Ileum;

B) Appendix;

✓ C) Duodenum ;

D) Jejunum;

E) ■olon.

Q200. In the histological preparation, the parenchyma of the organ is represented by lobules that have the shape of hexagonal prisms and consist of anastomosing plates, between which lie sinusoidal capillaries radially converging to the central vein. Determine which anatomical organ has this morphological structure?

A) thymus;

B) pancreas;

✓ C) liver;

D) spleen;

E) lymph node.

Q201. A patient, 30 years old, went to the doctor with complaints of an increase in body temperature to thirty-eight degrees, weakness, sore throat. During the examination, it turned out that the patient's tongue was covered with a white coating. Specify which histological structures of the tongue are involved in the formation of this plaque?

A) The epithelium of the fungal papillae;

B) Epithelium of the leaf-like papillae;

✓ C) Epithelium of filiform papillae;

D) The epithelium of the grooved papillae;

E) Connective tissue basis of all papillae of the tongue.

Q202. A gland consisting of several secretory compartments in the form of sacs, which open into one common excretory duct, was found in the micropreparation. What kind of gland is this?

✓ A) Simple branched alveolar;

B) Simple unbranched alveolar;

C) Simple branched tubular;

D) Complex unbranched alveolar;

E) Complex branched alveolar.

Q203. In the histological preparation, the mucous membrane is determined, covered with a multi-layered flat non–keratinizing, in places - a multi-layered flat keratinizing epithelium. The composition of the mucous membrane also includes its own plate, there is no muscle plate. Determine the location of such a mucous membrane?

A) Small intestine;

B) Esophagus;

✓ C) Oral cavity;

D) Trachea;

E) Stomach.

Q204. In the histological preparation, an organ is determined, the basis of which is skeletal striated muscle tissue. The organ has cutaneous, intermediate and mucous divisions. Epithelium – multilayered flat keratinizing in the mucous department turns into a multilayered flat non-keratinizing. Name this body?

A) Tongue;

✓ B) Lip;

C) Cheek;

D) Hard palate;

E) Gum.

Q205. Select, the muscle plate of the mucous membrane is determined in:

A) lip

B) cheek

C) gum

✓ D) esophagus

E) tongue

Q206. Specify what secret the parotid gland secretes?

A) muco-protein

✓ B) protein

C) protein-mucous

D) slimy

E) greasy

Q207. Determine whether the mobility of the mucous membrane on the lower surface of the tongue is provided by:

A) the epithelium of the mucous membrane

B) own record

C) muscle plate

✓ D) submucosal base

E) muscle membrane

Q208. Specify whether the bodies of odontoblasts are located in:

A) dentine

B) predentine

C) cement

✓ D) pulp

E) enamels

Q209. Specify where the esophageal glands are located?

A) mucosal epithelium

B) own plate of the mucous membrane

C) muscle membrane

✓ D) submucosal basis

E) adventitious shell

Q210. Explain how the serous membrane differs from the adventitious one:

A) the absence of blood vessels

B) the presence of nerve elements

C) lack of glands

✓ D) the presence of mesothelium

E) an abundance of adipose tissue

Q211. Choose in which part of the intestine the glands meet in the submucosal base:

A) the vermiform process

B) the colon

C) jejunum

✓ D) duodenum

E) the ileum

Q212. Specify which cells in the glands of the stomach produce pepsinogen:

✓ A) the main

B) parietal

C) neck

D) endocrine

E) mucocytes

Q213. Indicate whether the cambial cells in the epithelium of the small intestine are:

A) edged enterocytes of villi

✓ B) capless crypt enterocytes

C) goblet-shaped enterocytes

D) apical-granular enterocytes

E) endocrinocytes

Q214. List, smooth myocytes in the muscular lining of the stomach form:

A) one longitudinal layer

B) one transverse layer

C) two layers — longitudinal and transverse

✓ D) three layers - longitudinal, transverse and oblique

E) four layers alternating longitudinally and transversely

Q215. Give a definition. The villi of the small intestine are:

- ✓ **A) outgrowths of the mucous membrane**
- B) outgrowths of the integumentary epithelium
- C) a set of microvilli
- D) folds of the mucous and submucosal membranes

Q216. Indicate the source of the development of the epithelial lining of the stomach is:

- A) ectoderm
- ✓ **B) endoderm of the intestinal tube**
- C) mesoderm
- D) mesenchyme
- E) chorion

Q217. Classify the glands of the bottom of the stomach:

- A) simple branched alveolar
- ✓ **B) simple tubular unbranched**
- C) complex branched tubular
- D) simple unbranched alveolar
- E) complex unbranched tubular

Q218. Determine the distinctive features of the jejunum

- ✓ **A) villi**
- B) complex glands in the own plate of the mucosa
- C) crypts
- D) pits

Q219. Choose, intestinal peristalsis is caused by:

- A) movement of the villi
- ✓ **B) contractions of the muscular membrane**
- C) the presence of folds
- D) signals from the musculoskeletal plexus

Q220. Specify where in the hepatic lobules the perisinusoidal (Disce) space is located

- A) between hepatic beams
- B) inside the beams
- C) between hepatocytes
- ✓ **D) between sinusoid capillaries and beams**

Q221. Identify what pancreatic islet D-cells produce

A) insulin

B) glucagon

✓ C) somatostatin

D) vasoactive polypeptide

Q222. Indicate whether the presence of white plaque on the tongue is associated with impaired rejection of horny scales on the apex of

A) foliate papillae

B) fungiform papillae

✓ C) filiform papillae

D) circumvallate papillae

Q223. Identify the type of esophageal glands

A) simple branched alveolar mucous membranes

✓ B) complex branched alveolar-tubular mucous glands

C) complex branched alveolar-tubular proteinous

D) simple unbranched tubular protein

Q224. Specify which organ is characterized by villi, crypts, glands in submucosa

A) esophagus

B) the stomach

✓ C) duodenum

D) jejunum

E) ileum

Q225. Choose what the mucous membrane consists of

A) epithelium, muscularis mucosae, submucosa

B) epithelium, lamina propria, adventitia

✓ C) epithelium, lamina proria, muscularis mucosae

D) epithelium, submucosa, adventitia

Q226. Cell which forms dentin is:

A) Ameloblast

✓ B) Odontoblast

C) Osteocyte

D) Osteoblast

Q227. Hardest material in the body is:

A) Dentin

✓ B) Enamel

C) Cementum

D) Pulp tissue

Q228. Mucous glands of esophagus are found in:

A) Lining epithelium

✓ B) Submucosa

C) Muscularis externa

D) Adventitia

Q229. Nerve plexus of muscularis externa is called as:

A) Meissner’s plexus

✓ B) Aurebach’s plexus

C) Barrett’s plexus

D) Brachial plexus

Q230. Following cells are found in small intestine, except:

A) Stem cells

B) Goblet cells

✓ C) Neck cells

D) Paneth cells

Q231. Cells which give beaded cells to the mucosa of the stomach are:

A) Stem cells

B) Chief cells

✓ C) Neck cells

D) Parietal cells

Q232. Cells which store vitamin A in liver are:

A) Hepatocytes

✓ B) Ito cells

C) Cholangiocytes

D) Kupffer cells RESPIRATORY SYSTEM. SKIN AND IT’S APPENDEGES

Q233. Indicate whether the embryonic source of lung development is ...

A) the dorsal wall of the primary intestine

✓ B) the ventral wall of the primary intestine

C) the parietal leaf of the splanchnotome

D) visceral splanchnotome leaf

E) ectoderm

Q234. Select, the epithelium of the tracheal mucosa ...

A) single-layer flat

B) multilayer flat

C) single-row prismatic

D) cubic

✓ E) pseudostratified ciliated

Q235. Identify which epithelial cells of the trachea and bronchi produce mucus?

A) secretory

B) bordered

✓ C) goblet-shaped

D) endocrine

E) basal

Q236. Specify whether goblet cells synthesize ...

A) surfactant components

✓ B) mucus

C) serotonin

D) dopamine

E) adrenaline

Q237. Lung of premature infant is presented on electronic photomicrography of biopsy material. Collapse of the alveolar wall caused by the deficiency of surfactant was revealed. Disfunction of what cells of the alveolar wall caused it?

A) Fibroblasts

✓ B) Alveocytes type II

C) Alveolar macrophages

D) Alveocytes type I

E) Secretory cells

Q238. A patient was admitted to the hospital with an asphyxia attack provoked by a spasm of smooth muscles of the respiratory tracts. This attack was mainly caused by alterations in the following parts of the airways:

A) Respiratory part

B) Small bronchi

C) Large bronchi

D) Median bronchi

✓ E) Terminal bronchioles

Q239. Explain, the terminal sections of which glands are located in the submucosal base of the trachea ...

A) protein

B) mucous membranes

C) endocrine

✓ D) protein-mucosal

E) synthesizing surfactant

Q240. Specify which of the bronchi contains glands and cartilage in the form of islands in its wall?

A) the main

B) bronchus of the 1st order

C) bronchus of the 2nd order

D) bronchus of the 3rd order

✓ E) small bronchus

Q241. Determine which sections of the airways are most capable of changing the lumen?

A) larynx

B) trachea

C) medium-caliber bronchus

✓ D) small-caliber bronchus

E) bronchiola

Q242. Specify which epithelium is lined with the mucosa of the terminal bronchiole?

A) single-layer flat

B) two-row prismatic

C) multi-row flickering

✓ D) simple cuboidal ciliated

E) single-row prismatic

Q243. An employee of chemical production after inhaling caustic vapors had the death of a part of the ciliated epithelial cells of the bronchi. Determine which cells will regenerate this epithelium?

A) ciliated cells;

B) b) non-ciliated cells;

C) c) goblet cells;

D) d) endocrine cells;

✓ E) e) basal cells;

Q244. In the histological preparation of the trachea, low oval or triangular cells are visible as part of the multi-row ciliated epithelium. They do not reach the apical surface of the epithelium with their apex, mitosis figures are visible in part of the cells. Tell me what function these cells perform?

✓ A) are a source of regeneration;

B) secrete mucus;

C) produce biologically active substances;

D) secrete surfactant;

E) are part of the mucociliary complex;

Q245. The patient was admitted to the department with an attack of suffocation caused by a spasm of the smooth muscles of the respiratory tract. What are the departments of the airways with which this attack is mainly associated?

A) respiratory department;

B) small-caliber bronchi;

C) large-caliber bronchi;

D) medium-caliber bronchi;

✓ E) terminal bronchioles;

Q246. The preparation contains a hollow organ. The mucous membrane is covered with a double-row ciliated epithelium, which turns into a single-row one. The muscular plate of the mucosa is well developed in relation to the thickness of the entire wall. There is no cartilage and glands. Determine which organ is represented in the preparation?

A) larynx;

B) bladder;

C) middle bronchus;

D) trachea;

✓ E) small bronchus;

Q247. As a result of the pathological process in the bronchi, desquamation of the epithelium occurs. Choose which cells will regenerate the bronchial epithelium?

A) ciliated cells;

B) b) insertion cells;

C) c) goblet cells;

D) d) endocrine cells;

✓ E) e) basal cells;

Q248. When examining a patient with diphtheria, changes in the soft palate and tongue were revealed. Which epithelium was injured at the same time?

✓ A) stratified squamous non-keratinizing

B) multilayer cubic;

C) multi-row ciliated;

D) multilayer flat keratinizing;

E) single-layer prismatic.

Q249. A patient with acute rhinitis has hyperemia and increased mucus formation in the nasal cavity. Which mucosal epithelial cells are more active?

A) ciliated cells;

B) microvilli cells;

✓ C) goblet cells;

D) endocrine cells;

E) basal cells;

Q250. Electronic microphotographs of biopsy material show the lung of a premature baby. The collapse of the alveolar wall was detected due to the absence of surfactant. Specify the dysfunction of which cells of the alveolar wall cause this picture?

A) fibroblasts;

✓ B) type II alveolocytes;

C) alveolar macrophages;

D) type I alveolocytes;

E) secretory cells;

Q251. It is known that an important component of the aerogematic barrier is the surfactant alveolar complex, which prevents alveolar subsidence during exhalation. Choose which alveolar cells synthesize phospholipids that go to build surfactant membranes?

A) respiratory cells;

✓ B) type II alveolocytes;

C) alveolar macrophages;

D) capillary endothelium;

E) bordered epithelial cells;

Q252. Electronic micrographs of biopsy material show structures that include surfactant, type I alveolocytes, basement membrane and fenestrated capillary endothelium. Determine which histo-hematic barrier in the human body these structures belong to?

✓ A) aerogematic;

B) hematothymus;

C) hematotesticular;

D) hematolytic;

E) blood-brain;

Q253. In the histological preparation of the airways, the integumentary epithelium contains ciliated and goblet-shaped cells that form a mucociliary complex. Specify which function belongs to this complex?

A) hormone secretion;

B) humidification of the air;

C) respiratory;

✓ D) air purification from dust particles;

E) air warming;

Q254. The electronic micrograph shows structures in the form of open bubbles, the inner surface of which is lined with a single-layer epithelium, which is formed by respiratory and secretory cells. What are these structures?

A) Bronchioles;

B) Alveolar passages;

C) Terminal bronchi;

✓ D) Alveoli;

E) Acinuses;

Q255. Determine if pulmonary acinus begins ...

A) terminal bronchiola

✓ B) respiratory bronchiola

C) alveolar course

D) small bronchus

E) alveolar sacs

Q256. List, pulmonary acinus form:

A) a group of terminal bronchioles

B) one terminal bronchiole and two respiratory

C) alveolar passages

D) vestibules and alveolar sacs,

✓ E) respiratory bronchioles, alveolar passages and alveolar sacs

Q257. Choose which cells produce surfactant?

A) Type 1 alveocytes

B) endocrine cells

✓ C) Type 2 alveocytes

D) macrophages

E) goblet-shaped

Q258. Find out which structure of the respiratory system is lined with mesothelium?

A) larynx

B) trachea

C) bronchi

✓ D) pleura

E) lungs

Q259. Specify which epithelial cells of the bronchial mucosa produce a mucous secret rich in hyaluronic and sialic acids?

A) basal

B) endocrine

✓ C) goblet-shaped

D) bordered

E) unbranched

Q260. Choose which cells are found in the epithelium of the bronchial mucosa, but are not present in the epithelium of the nasal cavity

A) ciliated

B) endocrine

✓ C) Clara cells

D) basal

E) goblet-shaped

Q261. What is the epithelium covering the lungs from the outside?

✓ A) mesothelium

B) ciliated epithelium

C) connective tissue capsule

D) multilayer squamous epithelium

E) elastic membrane

Q262. Specify, the epiglottis mucosa is lined with epithelium:

A) single-layer flat

B) stratified squamous

C) multi-row ciliated

✓ D) multi-row on the larynx side and multi-layered on the pharynx side

E) transitional

Q263. Indicate if the paranasal sinuses are lined:

A) adventitious shell

B) periosteum

C) a mucous membrane with a multilayered squamous epithelium

✓ D) mucous membrane pseudostratified ciliated epithelium

E) serous membrane

Q264. Specify the cells involved in the secretion of surfactant components:

A) hemocapillary endotheliocytes

B) epithelial cells of terminal bronchioles

C) respiratory alveolocytes

✓ D) secretory alveolocytes

Q265. Choose whether the true vocal cords contain:

✓ A) stratified squamous epithelium

B) stratified epithelium

C) striated muscle tissue

D) bundles of smooth myocytes

Q266. Specify, respiratory bronchioles are lined with ...

A) stratified squamous keratinizing epithelium

B) stratified ciliated epithelium

✓ C) simple cuboidal epithelium

D) simple double-row epithelium

E) stratified squamous non keratinizing epithelium

Q267. In the interalveolar septa of the lungs are:

A) collagen fibers

B) elastic fibers

C) fibroblasts

D) macrophages

✓ E) everything is true

Q268. Choose which tissue forms the papillary layer of the skin?

A) dense connective tissue

✓ B) loose connective tissue

C) dense decorated

D) reticular

E) lymphoid

Q269. Specify, melanocytes are formed from ...

A) neural tube

B) ectodermal placodes

✓ C) neural crest

D) dermatoma

E) mesenchyma

Q270. Specify in which layer of the epidermis the cambial cells for keratinocytes are located?

A) granular

B) Brilliant

✓ C) basal

D) horny

E) spiky

Q271. Choose from the dermatome develop ...

A) skin epithelium

B) hair

C) mammary glands

✓ D) connective tissue of the skin

E) sebaceous glands

Q272. Determine whether melanocytes connect with each other using ...

A) synapses

B) nexuses

C) desmosis

D) half-cosmos

✓ E) lie freely

Q273. Name what hair develops from?

A) mesenchyma

B) the mesh layer of the dermis

C) the papillary layer of the dermis

✓ D) epidermis

E) connective tissue

Q274. Name which cells the epithelium of the sebaceous glands consists of?

A) myocytes

B) endocrinocytes

C) serocytes

D) mucocytes

✓ E) sebocytes

Q275. Specify in which layer of the skin are the end sections of the sebaceous glands located?

A) thorny

B) sprouts

C) basal

D) papillary

✓ E) reticular

Q276. Name, the growth of the hair occurs due to the cells ...

A) the cuticle of the hair

B) hair bag

✓ C) hair follicles

D) the brain matter of the hair

E) the cortical substance of the hair

Q277. Choose, the muscle that lifts the hair consists of ...

A) collagen fibers

B) elastic fibers

✓ C) smooth muscle cells

D) myofibrils

E) reticular fibers

Q278. Specify the type of sebaceous glands secreted?

A) autocrine

B) apocrine

C) paracrine

D) merocrine

✓ E) Holocrine

Q279. Which cells form skin pigment and give a positive reaction to DOPA oxidase?

A) Merkel cells

B) Langerhans cells,

C) keratinocytes

D) Schwann cells

✓ E) melanocytes

Q280. Choose which structures are missing in the skin?

A) sweat glands

B) melanocytes

C) sebaceous glands

D) papillary dermis layer

✓ E) photosensory layer of cells

Q281. Indicate whether the intermediate filaments of epithelial cells consist of ...

A) actin

✓ B) keratin

C) desmina

D) tubulin

E) vimentina

Q282. In the histological preparation of a biopsy of the epidermis of the skin of a healthy adult, dividing cells are visible in the basal layer. Choose which process these cells provide?

A) apoptosis;

✓ B) physiological regeneration;

C) adaptation;

D) differentiation;

E) reparative regeneration.

Q283. During wound healing, a connective tissue scar develops in the area of a skin tissue defect. Which cells provide this process?

A) melanocytes;

✓ B) fibroblasts;

C) fibrocytes;

D) macrophages;

E) mast cells.

Q284. An alien body got into the skin, which led to inflammation. Specify which connective tissue cells are involved in the skin's reaction to a foreign body?

A) melanocytes;

B) macrophages;

✓ C) neutrophils, macrophages, fibroblasts;

D) lipocytes;

E) adventitious cells.

Q285. The study of fingerprints of the epidermis of the fingers [fingerprinting] is used in criminology to identify individuals, as well as to diagnose genetic abnormalities, in particular Down's disease. Which layer of skin determines the individuality of the prints?

A) basal;

B) reticular;

✓ C) papillary;

D) shiny;

E) horny.

Q286. Patient A., 12 years old, has white spots on the skin that do not have pigment. Spots appeared after 10 years, constantly increasing in size. Name the absence of which skin cells led to the appearance of such spots?

A) Tissue basophils;

✓ B) Melanocytes;

C) Langerhans cells;

D) Keratinocytes;

E) Merkel Cells.

Q287. In the third week of embryogenesis, the central part of the epiblast cells (ectoderm) bends and the neurulation process begins. Determine in which direction the remaining cells, ectoderms, differentiate?

✓ A) Skin;

B) Somites;

C) Yolk bladder;

D) Chords;

E) Intestines.

Q288. On an electronic micrograph of the epidermis of the skin, process cells are distinguished among cubic cells, in the cytoplasm of which there is a well-developed Golgi apparatus, many ribosomes and melanosomes. Name this cell?

A) Tissue basophils;

✓ B) Melanocytes;

C) Langerhans cells;

D) Keratinocytes;

E) Merkel Cells.

Q289. Specify whether the trophic function of the skin is performed by a layer ...

A) granular

B) basal

C) spiky

✓ D) papillary

E) reticular

Q290. List which fabric forms the mesh layer of the skin…

✓ A) dense irregular

B) loose unformed

C) dense decorated

D) reticular

E) lymphoid

Q291. Choose, the cortical substance of the hair consists of:

A) polygonal cells with pigment granules

✓ B) flat horny scales with pigment granules

C) amorphous substance

D) cells of the germinal layer of the epidermis

E) dying melanocytes

Q292. Indicate whether hair growth occurs due to cell division:

A) brain matter.

B) cortical substance

C) the hair papilla

✓ D) hair follicles

E) hair follicle

Q293. Choose, the main tissue of the mesh layer of the dermis of the skin is:

A) loose connecting

B) dense decorated

✓ C) dense irregular

D) smooth muscle

E) striated muscle

Q294. Highlight, the growth of the nail plate occurs due to cell division:

A) its root

B) her body

✓ C) the nail matrix

D) epidermis of nail rollers

E) supra-elbow plate

Q295. Determine whether the center of the proliferative unit of the epidermis is considered:

A) keratinocyte of the basal layer in the interphase

✓ B) keratinocyte of the basal layer in mitosis

C) pigment cell

D) intraepidermal macrophage

E) Merkel's cage

Q296. In a fall, a child scraped the skin of his palm. Which epithelium was damaged in the fall?

✓ A) stratified squamous keratinizing epithelium

B) stratified ciliated epithelium

C) simple cuboidal epithelium

D) simple double-row epithelium

E) stratified squamous non keratinizing epithelium

Q297. A patient with a fractured index finger came to the trauma center. Select which of the injured tissues regenerates the fastest?

A) loose connecting tissue

B) dense regular connective tissue

C) dense connective tissue

✓ D) stratified squamous keratinizing epithelium

E) skeletal muscle tissue

Q298. Lining epithelium of alveoli is:

A) Simple columnar

✓ B) Simple squamous

C) Simple cuboidal

D) Pseudostratified columnar

Q299. Surfactant is secreted by:

A) Goblet cells

B) Type I pneumocytes

✓ C) Type II pneumocytes

D) Clara cells

Q300. Function of dust cells is:

✓ A) Phagocytosis

B) Secretion of surfactant

C) Exchange of gases

D) Absorption

Q301. Skeletal framework of epiglottis is formed by:

✓ A) Elastic cartilage

B) Hyaline cartilage

C) White fibrocartilage cartilage

D) Reticular tissue

Q302. Which of the following statements about the sebaceous glands is true?

A) They control the body temperature

✓ B) They are always associated with hair follicles

C) They are abundant on the face only

D) They are a source of new hairs

Q303. The dermis is made up of two layers:

A) Adipose tissue and basal layer

B) Papillary layer and horny layer

✓ C) Papillary and reticular layers

D) Inner layer and granular layer

Q304. Melanin protects the skin from:

A) Pathogenic bacteria

B) Electric heat

C) Steam heat

✓ D) Ultraviolet rays URINARY SYSTEM

Q305. A patient with suspected glomerulonephritis has the presence of albumins (albuminuria) and glucose (glucosuria) in the secondary urine for two weeks. Choose which parts of the kidney function is impaired?

A) juxtaglomerular apparatus;

✓ B) proximal tubules;

C) loop of Henle;

D) distal tubules;

E) collecting tubes.

Q306. Histological examination of the kidney in the cortical substance determines a tubule lined with a single-layer cubic edged epithelium, the cytoplasm of which is colored oxyphilically. Specify which segment of the nephron is detected in the drug?

A) distal rectus canaliculus;

✓ B) proximal convoluted canaliculus;

C) Henle loop;

D) distal convoluted tubule;

E) collecting tube.

Q307. An electron micrograph of a fragment of a renal corpuscle shows a large epithelial cell with large and small processes. The latter are attached to the basement membrane of the capillaries. Name this cell?

A) juxtavascular cell;

✓ B) podocyte;

C) endotheliocyte;

D) mesangial cell;

E) smooth myocyte.

Q308. An electron micrograph of a kidney fragment shows a bearing arteriole, in which large cells containing secretory granules are visible under the endothelium. Determine this type of cells?

A) an increase in the amount of testosterone;

B) aspermatogenesis;

✓ C) polyspermia;

D) monospermia;

E) a decrease in testosterone synthesis

Q309. Leached erythrocytes were found in the patient's urine. Name which part of the nephron is damaged?

✓ A) the membrane of the renal corpuscle;

B) proximal convoluted tubule;

C) Henle loop;

D) distal convoluted tubule;

E) collecting tube.

Q310. Cells with large secretory granules in the cytoplasm are determined on electronic micrographs of the kidney site in the wall of the bringing and taking out arterioles. Determine the structural formation of the kidney, which includes these cells?

A) the renal corpuscle;

B) the proximal nephron;

C) the Henle loop;

D) distal nephron;

✓ E) juxtaglomerular apparatus

Q311. Leached erythrocytes were found in the analysis of the patient's urine. Where is the localization of the pathological process possible?

✓ A) filtration barrier;

B) proximal convoluted tubule;

C) Henle loop;

D) distal convoluted tubule;

E) collecting tube.

Q312. A 50-year-old patient with chronic nephritis developed anemia. Determine what was the most likely cause of anemia in this patient?

✓ A) Decrease in erythropoietin production;

B) Violation of the synthesis of adrenaline;

C) Immunological damage to erythropoiesis progenitor cells;

D) Lack of iron;

E) Lack of vitamin B12.

Q313. Sugar was found in the urine of a 30-year-old patient with its normal amount in the blood. What structural and functional mechanisms of the kidney are damaged?

A) Filtration process;

B) The process of rearbsorption in a thin tubule;

C) The process of rearbsorption in the distal part as a result of insufficient secretion of ADH;

✓ D) The process of rearbsorption in the proximal nephron;

E) The process of rearbsorption in the distal part of the nephron.

Q314. During a clinical examination in a 35-year-old woman with kidney disease, blood cells, fibrinogen, were found in the urine, probably due to a violation of the renal filter. Determine what structures this filter consists of?

A) Capillary endothelium, basement membrane;

B) Three-layer basement membrane;

✓ C) Endothelium of glomerular capillaries, three-layer basement membrane, podocytes;

D) Podocytes, basal membrane;

E) Endothelium, podocytes.

Q315. Electron microscopy of the kidney revealed tubules lined with cubic epithelium. There are light and dark cells in the epithelium. There are few organelles in light cells. The cytoplasm forms folds. These cells ensure the reabsorption of water from the primary urine into the blood. Dark cells resemble the parietal cells of the stomach in structure and function. Specify which tubules are represented on the electronogram?

A) filtration barrier;

B) proximal convoluted tubule;

C) Henle loop;

D) distal convoluted tubule;

✓ E) collecting tube.

Q316. Specify which endocrine cells in the kidney secrete renin?

A) interstitial

B) mesangial cell

✓ C) juxtaglomerular

D) podocytes

E) macula dense

Q317. Blood in the renal arcuate arteries flows next into which vessels?

A) a) Afferent arterioles

B) b) Efferent arterioles

C) c) Glomerular capillaries

D) d) Interlobar arteries

✓ E) e) Interlobular arteries

Q318. Which cell type comprises the visceral layer of Bowman capsule?

A) a) Endothelial cells

B) b) Juxglomerular cells

C) c) Mesangial cells

✓ D) d) Podocytes

E) e) Extraglomerular mesangial (or Lacis) cells

Q319. Which type of epithelium lines the thick ascending limb of the loop of Henle?

A) a) Pseudostratified columnar

B) b) Simple columnar

✓ C) c) Simple cuboidal

D) d) Simple squamous

E) e) Transitional (urothelium)

Q320. Which cell is a modified smooth muscle cell that secretes rennin?

A) a) Macula densa cells

B) b) Mesangial cells

C) c) Podocytes

✓ D) d) Juxtaglomerular cells

E) e) Endothelial cells

Q321. Epithelial cell membrane domains containing many stiffened plaques of protein are an important feature in which part of the urinary system?

A) a) Juxtaglomerular apparatus

✓ B) b) Bladder mucosa

C) c) Collecting ducts

D) d) Renal pyramids

E) e) Membranous urethra

Q322. An immunohistochemical technique using antibodies against aquaporins to stain a section of kidney would be expected to stain cells in which structures most intensely?

A) a) Collecting ducts

B) b) Lining of major and minor calyces

✓ C) c) Proximal convoluted tubules

D) d) Distal convoluted tubules

E) e) Glomeruli

Q323. What type of epithelium lines the prostatic urethra?

A) a) Simple columnar

B) b) Pseudostratified columnar

C) c) Stratified squamous

D) d) Simple squamous

✓ E) e) Transitional (urothelium)

Q324. A 45-year-old man presents with nephrolithiasis or kidney stones. The process of calcium oxalate stone formation as seen in this patient begins with Randall plaques found in the basement membranes of which one of the following structures found only in the renal medulla?

A) a) Proximal convoluted tubules

B) b) Distal convoluted tubules

✓ C) c) Thin loops of Henle

D) d) Afferent arterioles

E) e) Collecting ducts

Q325. A 15-year-old male presents with hematuria, hearing loss, lens dislocation, and the onset of cataracts. Genetic analysis reveals a mutation in the COL4A5 gene. Transmission EM examination of a renal biopsy confirms that the disorder has affected a component of the renal corpuscles in which damage disrupts normal glomerular filtration. Which one of the following structures would most likely be abnormal in the TEM of this patients biopsy?

A) Pedicels

B) Filtration slits

C) Slit diaphragms

✓ D) Glomerular basement membranes

E) Fenestrated endothelium of glomerular capillaries

Q326. Specify where mesangiacytes are located in the kidneys

A) in the inner leaflet of the glomerulus capsule

B) as a part of macula densa

C) next to intercanal capillaries

✓ D) between capillaries of glomerulus vascularis

E) around of afferent and efferent arterioles

Q327. What makes the renal corpuscle?

✓ A) Bowman’s capsule and glomerulus

B) Bowman’s capsule and tubules

C) Glomerulus and tubules

D) Bowman’s capsule, glomerulus, and tubules

Q328. Renal corpuscles are found in:

✓ A) Cortex

B) Medulla

C) Both cortex and medulla

B) Pyramid

Q329. Podocytes are the cells that lines the:

A) Glomerulus

✓ B) Bowman’s capsule

C) Tubules

D) Ducts

Q330. What is the function of the brush border on the proximal convoluted tubule?

A) Decrease the surface area

B) Secretion

C) Motility

✓ D) Absorption

Q331. Renin is secreted by which cell?

A) Macula densa

✓ B) Juxtaglomerular

C) Interstitial

D) Lacis

Q332. Ureter is lined by ______________ epithelium.

A) Squamous

B) Stratified squamous

✓ C) Transitional

D) Pseudostratified REPRODUCTIVE SYSTEM

Q333. Specify the source of testicular development ...

A) the epithelium of the paramesonephral ducts

✓ B) thickening of the coelomic epithelium of the primary kidneys

C) the epithelium of the mesonephral ducts

D) the epithelium of the pariental leaf of the splanchnotome

E) coelomic epithelium

Q334. Determine whether the primary germ cells are formed for the first time ...

A) in the mesenchyma of the trunk

B) in somites

✓ C) in the wall of the yolk sac

D) in the primary kidney

E) in the amnion

Q335. Specify what does not develop from the wolf ducts?

A) appendages of the testis

✓ B) bulbourethral glands

C) seminal vesicles

D) Vas deferens

E) testicular network

Q336. Specify the structure from which the scrotum develops ...

A) genitourinary sinus

✓ B) floor rollers

C) sexual tubercle

D) wolf's duct

E) Muller's channel

Q337. What does not characterize the development of male sexual structures?

A) in the rudiments of the male sex glands the brain substance receives predominant development

B) The Y chromosome controls the critical stage of sexual differentiation...

C) primary germ cells differentiate in spermatogony

✓ D) under the action of the Muller inhibitory factor, the wolf ducts they differentiate into male sexual structures

E) primary germ cells migrate from the wall of the yolk sac into the genital rollers

Q338. Choose what does not characterize spermatogenesis?

A) lasts 65 days

B) occurs at a temperature below body temperature

✓ C) consists of the stage of reproduction and maturation

D) begins with the onset of puberty

E) occurs in the convoluted seminal tubules

Q339. Specify which of the functions are performed by the testes?

A) generative and immune

✓ B) generative and endocrine

C) endocrine and excretory

D) immune and secretory

E) hematopoietic and endocrine

Q340. Name the type of tissue that is formed by the stroma of the testis ...

A) dense connective tissue

✓ B) loose connective tissue

C) reticular tissue

D) epithelial

E) muscular

Q341. Specify the path of movement of spermatozoa in the genital tract of men ...

A) straight tubules - seminal tubules - convoluted excretory tubules - duct of the appendage - vas deferens - ejaculatory canal

B) convoluted outflow tubules - seminal tubules - straight tubules - duct of the appendage - ejaculatory canal

✓ C) seminal tubules - straight tubules - convoluted outflow tubules - duct of the appendage - vas deferens - vas deferens

D) seminal tubules - convoluted outflow tubules - straight tubules - ejaculatory canal

E) seminal tubules - straight tubules - duct of the appendage - vas deferens - vas deferens

Q342. Determine which cells are located between the seminal tubules and synthesize androgens?

A) supporting cells

B) fibroblasts

C) adventitious cells

✓ D) interstitial cells

E) reticular cells

Q343. Specify which hormone is synthesized by interstitial testicular cells?

A) estrogen

B) insulin

C) thyroxine

D) lutropin

✓ E) Testosterone

Q344. Determine which of the functions are not performed by the supporting cells of the testis?

A) provide nutrition to developing germ cells

B) protect developing germ cells from harmful effects

C) phagocytic degenerating germ cells and residual bodies

✓ D) synthesize follicle-stimulating hormone

E) synthesize androgen-binding hormone

Q345. Give a definition. The hematotesticular barrier is...

✓ A) capillary endotheliocyte - endothelial basement membrane - interstitial connective tissue - myoid cell layer - tubule basement membrane - dense junctions of sustentocytes

B) the basement membrane of the endothelium - capillary endotheliocyte - a layer of myoid cells - dense junctions of sustentocytes

C) capillary endotheliocyte - tubule basement membrane - dense junctions of sustentocytes

D) interstitial tissue - capillary endotheliocyte - myoid cell layer - dense junctions of sustentocytes

E) capillary endotheliocyte - capillary basement membrane -dense junctions of sustentocytes

Q346. List, spermatogenesis successively goes through the following phases ...

A) formation - reproduction - growth - maturation

✓ B) reproduction - growth - maturation - formation

C) growth - reproduction - maturation - formation

D) maturation - growth - reproduction - formation

E) maturation - reproduction - growth formation

Q347. Choose which functions are performed by Leydig cells?

A) synthesize luteinizing hormone

✓ B) synthesize testosterone

C) protect the germ cells from harmful effects

D) serve as supporting elements for germ cells

E) synthesize estrogen

Q348. Specify when spermatogonia enter the stage of reproduction?

A) at the 4th month of intrauterine development

✓ B) with the onset of puberty

C) immediately after differentiation from primary germ cells

D) after lowering the testicles into the scrotum

E) during the period from birth to puberty

Q349. Specify whether the spermatozoa from the straight tubules fall into ...

A) the carrying tubules

B) ejaculatory duct

✓ C) testicular network

D) the duct of the appendage

E) ampoule of the vas deferens

Q350. Choose which cells synthesize the Muller inhibitory factor?

A) interstitial cells

✓ B) sustentocytes

C) primary germ cells

D) pituitary gonadotropocytes

E) pituitary somatotropocytes

Q351. In case of mechanical injury of the testis in a man, a violation of the integrity of the walls of many tubules was noted. Choose what it will lead to?

✓ A) Aspermatogenesis;

B) An increase in the amount of testosterone;

C) Reduction of testosterone synthesis;

D) Monospermia;

E) Polyspermia.

Q352. In the study of familial fluid in a patient aged 25 years, an insufficient number of germ cells was revealed. Which of the cells of the male sex glands, dividing, provide a sufficient number of spermatozoa for fertilization?

✓ A) Spermatogonia;

B) Supporting cells;

C) Leydig cells;

D) Sertoli Cells;

E) Sustentocytes

Q353. In case of mechanical injury of the scrotum, violations of the epithelial lined network of the testis were revealed in the patient. Specify which epithelium was injured?

✓ A) Simple cuboidal;

B) Single-layer prismatic;

C) Transitional;

D) Double-row;

E) Multilayer flat.

Q354. Choose which cells secrete androgen binding protein?

✓ A) Sertoli cells

B) Leydig cells

C) seminal vesicle cells

D) prostate cells

E) bulbourethral gland cells

Q355. Specify the epithelium, which consists of high cylindrical cells with cilia, and low cubic cells with microvilli lining ...

A) the vas deferens

B) straight tubules

C) ovarian network

✓ D) the duct of the appendage

E) urethra

Q356. Choose, the formation of male germ cells occurs in

A) straight tubules of the testis

B) tubules of the testis network

✓ C) convoluted tubules of the testis

D) the outputting tubules of the testis

E) the duct of the appendage

Q357. Specify the sources of ovarian development ...

✓ A) thickening of the coelomic epithelium of the primary kidneys

B) outgrowths of the genitourinary sinus

C) the epithelium of the paramesonephral duct

D) coelomic epithelium of the dorsal mesentery

E) epithelium of the parietal leaf of the splanchnotome

Q358. Choose, ovogonia are formed in ...

✓ A) the ovary of the embryo

B) the ovary of an adult woman

C) the period of maturation

D) oviduct

E) the appendage of the ovary

Q359. Specify the sources of development of the fallopian tubes ...

A) the upper part of the mesonephral duct

✓ B) paramesonephral ducts

C) genitourinary sinus

D) metanephridia

E) segmental legs of somites

Q360. Give a definition. Mammary glands are modified ...

A) sebaceous glands

✓ B) skin sweat glands

C) digestive glands

D) salivary glands

E) lacrimal glands

Q361. Determine whether the primordial follicle consists of…

A) ovocyte, transparent shell, cylindrical follicular cells

✓ B) an ovocyte, a single layer of flat follicular cells

C) ovocyte, egg-bearing tubercle, internal and external fluid

D) ovocyte. radiant crown, follicular fluid

E) ovocyte, follicular fluid, external fluid

Q362. List, the primary follicle consists of…

A) ovocyte, a single layer of flat follicular cells

✓ B) ovocyte, transparent shell, 2-3 layers of cylindrical follicular cells

C) ovocyte, egg-bearing tubercle, internal fluid

D) ovocyte, radiant crown, follicular fluid, external fluid,

E) ovocyte, follicular fluid, internal and external fluid

Q363. The ovarian cycle includes the following sequence of events ...

A) follicle growth - formation of the corpus luteum - activity of the corpus luteum - its regression

✓ B) follicle growth - ovulation - formation and active function of the corpus luteum - its regression - growth of a new follicle

C) ovulation - follicle growth - formation and functioning of the corpus luteum

D) follicle growth - ovulation - growth of a new follicle

E) follicle growth - ovulation - follicle regression - growth of a new follicle

Q364. Indicate whether the uterine myometrium is formed by ...

A) striated muscle tissue.

✓ B) smooth muscle tissue

C) cardiac striated muscle tissue

D) myoepithelial cells

E) pericytes

Q365. Determine which hormone causes the synthesis of estrogens?

A) estrogen

B) lutropin

✓ C) follitropin

D) thymosin

E) thyroid-stimulating hormone

Q366. Determine which hormone causes the synthesis of progesterone?

✓ A) lutropin

B) follitropin

C) oxytocin

D) relaxin

E) inhibin

Q367. Specify which cells secrete estrogen?

A) neurosecretory cells of the hypothalamus

B) gonadotropocytes of the adenohypophysis

✓ C) cells of the inner shell of the secondary follicle

D) somatotropocytes

E) interstitial testicular cells

Q368. Choose which cells secrete progesterone?

A) gonadotropocytes of the adenohypophysis

B) interstitial testicular cells

C) ovocytes

✓ D) luteal cells of the corpus luteum

E) chromaffin cells of the adrenal glands

Q369. Classify, the uterine glands belong to…

A) simple alveolar

✓ B) simple tubular

C) complex alveolar with branched ends

D) complex tubular with branched ends

E) complex alveolar-tubular

Q370. Specify, after ovulation, the follicle is formed in place of:

A) white body

✓ B) yellow body

C) atretic body

D) mature follicle

E) growing follicle

Q371. With cyclic changes of the uterus, the most pronounced morphological rearrangement is subjected to:

A) myometrium

B) the basal layer of the endometrium

✓ C) functional layer of the endometrium

D) perimeters

E) the entire wall of the organ

Q372. Mass atresia of ovarian follicles, accompanied by estrogenization of the body, occurs during:

A) embryonic

B) prepubescent

C) pregnancy

✓ D) climacteric

E) senile

Q373. Specify whether the intra-follicular fluid in the ovary is secreted:

A) ovogonia

B) ovocyte of the first order

C) ovocyte of the II order

✓ D) follicular cells

E) interstitial cells

Q374. Gonoblasts, progenitors of germ cells, were detected in the embryo for 2-3 weeks. In what material do these cells differentiate?

✓ A) in the yolk sac;

B) in the rudimentary ectoderm;

C) in the rudimentary endoderm;

D) in the dermatome;

E) in the mesenchyme;

Q375. In the ovarian preparation, along with follicles of different orders, there are atretic bodies and a developed yellow body. Specify which stage of the ovarian-mental cycle corresponds to such a state in the ovary?

A) regenerative;

B) follicle growth;

C) menstrual;

D) postmenstrual;

✓ E) premenstrual;

Q376. In the histopreparation of a woman's ovary, a rounded formation consisting of large glandular cells containing the pigment lutein is revealed. In the center of this structure there is a small connective tissue scar. Specify the structure of the ovary?

A) primary follicle;

B) primordial follicle;

C) mature follicle;

✓ D) yellow body;

E) atretic body;

Q377. Normal implantation of a human embryo can only take place with a corresponding change in the endometrium of the uterus. Find out which endometrial cells are quantitatively increased at the same time?

A) myocytes;

B) neurons;

C) fibroblasts;

✓ D) decidual cells;

E) macrophages;

Q378. An increased amount of estrogens was found in the woman's blood. Specify which ovarian cells are involved in the formation of these hormones?

✓ A) Interstitial and follicular cells of secondary follicles;

B) Follicular cells of primary follicles;

C) Follicular cells and ovocytes;

D) Follicular cells of primordial follicles;

E) Ovocytes;

Q379. Irregular shapes of bright pink color (stained with hematoxylin and eosin) are observed on a slice of a normal ovary. As a result, what formed these figures?

A) White body formation;

B) Necrosis of the follicle;

C) Formation of the corpus luteum;

D) Ovulation;

✓ E) Follicle atresia;

Q380. A blood test of a non-pregnant woman at the age of 26 revealed a low concentration of estrogens and high progesterone. Determine at what stage of the ovarian-menstrual cycle was the analysis done?

A) Menstrual phase;

B) Desquamation phase;

C) Endometrial proliferation phase;

✓ D) Premenstrual phase (secretory);

E) Postmenstrual phase (proliferative);

Q381. A patient with pituitary adenoma (neoplasms in the anterior pituitary gland) has an increase in the duration of the phase of large follicle growth. What is the duration of the period of large growth of ovocytes in the process of ovogenesis normally?

A) 28 days;

B) 12-14 days;

C) After birth and before puberty;

✓ D) Several years (from 10-13 to 40-50) after birth;

E) From 3 months of prenatal development until birth;

Q382. A human embryo was found in the uterine cavity, not attached to the endometrium. What stage of development does the embryo correspond to?

A) Zygotes;

B) Gastrules;

C) Neurules;

✓ D) Blastocysts;

E) Morules;

Q383. The preparation of the uterine endometrium is presented. On the preparation, the endometrium is covered with a high ciliated epithelium, branched glands, many decidual cells. Indicate which stages of the menstrual cycle demonstrate this drug?

A) menstrual;

B) premenstrual;

C) postmenstrual;

✓ D) pregnancy;

Q384. Choose, lactating mammary glands are:

A) simple tubular

B) simple alveolar

✓ C) complex alveolar

D) complex tubular

E) unbranched

Q385. In the ovary specimen colored with hematoxylin-eosin, follicle is determined where cubic-shaped follicle epithelium cells are placed in 1-2 layers, and scarlet covering is seen around ovocyte. Name this follicle:

A) a) Atretic

✓ B) b) Primary

C) c) Secondary

D) d) Primordial

E) e) Mature

Q386. An ovary specimen stained by hematoxylin-eosin presents a follicle, where cells of follicular epithelium are placed in 1-2 layers and have cubic form, there is a bright-red membrane around the ovocyte. What follicle is it?

A) a) Secondary

B) b) Primordial

✓ C) c) Primary

D) d) Mature

E) e) Atretic

Q387. Which of the following is characteristic of granulosa lutein cells?

A) Are a minor cell type in the corpus luteum

B) Derive from the theca interna

C) Contain abundant rough endoplasmic

D) Are small and dark-staining

✓ E) Secrete progesterone

Q388. Which of the following hormones is primarily responsible for inducing ovulation?

A) Relaxin

✓ B) Luteinizing hormone

C) Progesterone

D) Follicle-stimulating hormone

E) Estrogen

Q389. Endometrial glands are typically most fully developed and filled with product during which day(s) or phase of a woman's menstrual cycle?

A) Menstrual phase

B) Day 1-4

C) The day ovulation occurs

D) Proliferative phase

✓ E) Day 15-28

Q390. What is the most commonly seen type of epithelium in the prostate?

A) Transitional

B) Pseudostratified squamous

C) Simple squamous

✓ D) Simple cuboidal

Q391. What type of gland composes the prostate?

A) Simple tubular gland

B) Simple alveolar gland

C) Compound tubular gland

✓ D) Compound tubuloalveolar gland

Q392. What constitutes the stroma of the prostate?

A) Loose irregular connective tissue

B) Smooth muscle

✓ C) Fibromuscular

D) Adipose tissue

Q393. How many seminiferous tubules are found in each testis of an average man?

A) 4–6

B) 40–60

✓ C) 400–600

D) 4,000–6,000

Q394. What are developing gametes called?

A) Oogenesis

B) Ovary

✓ C) Oocyte

D) Ova

Q395. Which stage of the follicle is arrested in prophase?

✓ A) Primordial follicle

B) Primary follicle

C) Secondary follicle

B) Mature follicle

Q396. What is the cavity within a secondary follicle?

A) Graafian follicle

B) Theca folliculi

C) Zona pellucida

✓ D) Antrum

Q397. Which of these secrete androgens in corpus luteum?

A) Connective septa

B) Granulosa-lutein cells

✓ C) Theca lutein cells

D) Corpus albicans

Q398. A histological section of the cerebellar cortex demonstrates complete degeneration of Purkinje cells. Which pathway is most directly interrupted?

✓ A) Transmission of impulses from climbing and mossy fibers to deep cerebellar nuclei

B) Sympathetic innervation of blood vessels

C) Formation of cerebrospinal fluid

D) Olfactory signal conduction

Q399. A patient develops inability to perform precise voluntary movements after damage to Betz cells. In which cortical layer are these neurons located?

A) External granular layer

B) Molecular layer

✓ C) Internal pyramidal layer

D) Multiform layer

Q400. During retinal examination, destruction of the outer nuclear layer is detected. Which cells are primarily located in this layer?

A) Ganglion cells

B) Bipolar neurons

✓ C) Photoreceptor cells

D) Horizontal cells

Q401. A patient loses the ability to detect odors after trauma to the cribriform plate. Which cells of the olfactory epithelium are directly responsible for odor reception?

A) Supporting cells

B) Basal cells

✓ C) Olfactory receptor neurons

D) Goblet cells

Q402. Histological analysis of the cochlea reveals degeneration of stereocilia-containing receptor cells. Which embryologic structure gives rise to these cells?

A) Neural crest

✓ B) Surface ectoderm

C) Endoderm

D) Mesoderm

Q403. A patient with severe hypertension has hypertrophy of smooth muscle in arteries. Which layer of the vessel wall is predominantly affected?

A) Tunica intima

B) Tunica adventitia

✓ C) Tunica media

D) Subendothelial layer

Q404. A vessel biopsy demonstrates numerous elastic lamellae between smooth muscle cells. Which vessel is this most likely?

A) Venule

B) Muscular artery

✓ C) Elastic artery

D) Capillary

Q405. A lymph node biopsy reveals absence of germinal centers after radiation exposure. Which cells are unable to proliferate?

A) T-helper lymphocytes

B) Plasma cells

✓ C) Activated B lymphocytes

D) Macrophages

Q406. Histological examination of the thymus shows absence of Hassall corpuscles. Which region is affected?

A) Cortex

B) Capsule

✓ C) Medulla

D) Trabeculae

Q407. Bone marrow biopsy demonstrates marked reduction in erythropoiesis. Which cells normally produce erythrocytes?

A) Megakaryocytes

✓ B) Erythroblasts

C) Osteoclasts

D) Reticular cells

Q408. A patient with medullary thyroid carcinoma has proliferation of parafollicular cells. Which hormone is secreted in excess?

A) Thyroxine

B) Triiodothyronine

✓ C) Calcitonin

D) Parathyroid hormone

Q409. Histological analysis of the adrenal gland reveals destruction of lipid-rich spongiocytes. Which hormone secretion is most affected?

A) Epinephrine

✓ B) Cortisol

C) Calcitonin

D) Thyroxine

Q410. A premature newborn develops respiratory distress syndrome due to surfactant deficiency. Which cells failed to mature?

A) Type I pneumocytes

✓ B) Type II pneumocytes

C) Alveolar macrophages

D) Clara cells

Q411. Chronic smoking causes squamous metaplasia of the tracheal epithelium. Which normal epithelium is replaced?

A) Transitional epithelium

B) Simple cuboidal epithelium

✓ C) Pseudostratified ciliated columnar epithelium

D) Stratified cuboidal epithelium

Q412. Histological examination of skin demonstrates absence of the stratum granulosum. From which region was the specimen most likely taken?

A) Thick skin of palm

B) Thick skin of sole

✓ C) Thin skin

D) Lip vermilion border

Q413. Gastric biopsy demonstrates autoimmune destruction of parietal cells. Which complication is most likely?

A) Iron deficiency from reduced ferritin synthesis

✓ B) Pernicious anemia due to intrinsic factor deficiency

C) Hyperglycemia due to insulin deficiency

D) Reduced bile production

Q414. Histological examination of the duodenum demonstrates severe damage to Brunner glands. Which function is primarily impaired?

A) Absorption of lipids

✓ B) Neutralization of acidic chyme

C) Production of digestive enzymes

D) Reabsorption of bile salts

Q415. A patient with Crohn disease has inflammation involving Peyer patches. In which intestinal segment are these structures most abundant?

A) Duodenum

B) Jejunum

✓ C) Ileum

D) Colon

Q416. Liver biopsy demonstrates disruption of hepatic plates and sinusoidal dilation. Which cells lining the sinusoids are specialized macrophages?

A) Ito cells

✓ B) Kupffer cells

C) Cholangiocytes

D) Hepatocytes

Q417. Kidney biopsy reveals destruction of the macula densa. Which renal function is most impaired?

A) Reabsorption of glucose

✓ B) Regulation of glomerular filtration rate

C) Secretion of erythropoietin

D) Concentration of urine

Q418. A patient with cerebellar damage cannot maintain posture and balance. Which neurons form the only output pathway from the cerebellar cortex?

A) Granule cells

B) Basket cells

✓ C) Purkinje cells

D) Stellate cells

Q419. Histological examination of the cerebral cortex shows destruction of pyramidal neurons. Which function is mainly impaired?

A) Sensory reception

✓ B) Voluntary motor activity

C) Hormone secretion

D) Memory storage

Q420. A patient develops color blindness due to retinal degeneration. Which cells are primarily damaged?

A) Rods

✓ B) Cones

C) Bipolar cells

D) Ganglion cells

Q421. Which cells of the olfactory epithelium are capable of regeneration after injury?

A) Supporting cells

B) Goblet cells

✓ C) Basal cells

D) Olfactory neurons

Q422. A patient with hearing loss has degeneration of receptor cells in the cochlea. Where are these cells located?

A) Spiral ganglion

✓ B) Organ of Corti

C) Semicircular canals

D) Tympanic membrane

Q423. Histological examination reveals a vessel with one endothelial layer only. Which vessel is this?

A) Elastic artery

B) Vein

✓ C) Capillary

D) Muscular artery

Q424. A patient has varicose veins due to valve insufficiency. Which layer forms venous valves?

A) Tunica media

B) Tunica adventitia

✓ C) Tunica intima

D) Elastic membrane

Q425. A biopsy of the spleen shows destruction of red pulp. Which process is most impaired?

A) T-cell maturation

B) Antibody synthesis

✓ C) Destruction of old erythrocytes

D) Platelet formation

Q426. Histological examination of lymph node cortex reveals follicles with germinal centers. Which cells predominate there?

A) T lymphocytes

✓ B) B lymphocytes

C) Macrophages

D) Plasma cells

Q427. A patient with pituitary adenoma secretes excess growth hormone. Which cells are responsible?

A) Basophils

B) Chromophobes

✓ C) Acidophils

D) Pituicytes

Q428. A patient with hypocalcemia has low parathyroid hormone secretion. Which cells normally produce this hormone?

A) Oxyphil cells

B) Follicular cells

✓ C) Chief cells

D) Chromaffin cells

Q429. Histological analysis of adrenal medulla reveals destruction of chromaffin cells. Which hormone secretion decreases?

A) Cortisol

B) Aldosterone

✓ C) Epinephrine

D) Thyroxine

Q430. Chronic smoking damages cilia in the respiratory tract. What is the main function of cilia?

A) Hormone secretion

✓ B) Mucus transport

C) Gas exchange

D) Immune response

Q431. A newborn has respiratory distress because alveoli collapse during expiration. Which substance is deficient?

A) Mucin

B) Keratin

✓ C) Surfactant

D) Collagen

Q432. Histological examination of skin reveals absence of hair follicles and sebaceous glands. Which skin is this?

A) Thin skin

B) Scalp skin

✓ C) Thick skin

D) Axillary skin

Q433. A patient with severe burns loses thermoregulation. Which skin glands are most responsible for cooling the body?

A) Sebaceous glands

B) Apocrine glands

✓ C) Eccrine sweat glands

D) Ceruminous glands

Q434. Gastric biopsy shows destruction of chief cells. Which secretion decreases?

A) Hydrochloric acid

✓ B) Pepsinogen

C) Gastrin

D) Intrinsic factor

Q435. Histological examination shows intestinal villi lined by simple columnar epithelium with microvilli. What is the main function of microvilli?

A) Protection

B) Secretion

✓ C) Increase absorption surface

D) Hormone synthesis

Q436. A section of ileum shows aggregated lymphoid nodules. What are these structures called?

A) Crypts of Lieberkühn

B) Brunner glands

✓ C) Peyer patches

D) Lacteals

Q437. A patient with ulcerative colitis has severe goblet cell loss in the colon. Which function is impaired?

A) Protein digestion

B) Enzyme secretion

✓ C) Lubrication of feces

D) Fat absorption

Q438. Histological examination of the liver reveals fat accumulation in hepatocytes. Which organelle is primarily responsible for detoxification in these cells?

A) Golgi complex

B) Rough ER

✓ C) Smooth ER

D) Lysosomes

Q439. A patient with diabetes mellitus has destruction of pancreatic beta cells. Which hormone decreases?

A) Glucagon

✓ B) Insulin

C) Somatostatin

D) Secretin

Q440. Kidney biopsy shows damage to proximal convoluted tubules. Which process is mainly impaired?

A) Filtration

✓ B) Reabsorption

C) Urine storage

D) Hormone secretion

Q441. Histological examination of the urinary bladder reveals transitional epithelium. What is its major property?

A) Keratinization

B) Rapid mitosis

✓ C) Ability to stretch

D) Cilia formation

Q442. Testicular biopsy reveals absence of Leydig cells. Which hormone production decreases?

A) FSH

B) LH

✓ C) Testosterone

D) Estrogen

Q443. A patient with infertility has impaired sperm maturation in the epididymis. Which epithelium lines this organ?

A) Stratified squamous

B) Transitional

✓ C) Pseudostratified columnar with stereocilia

D) Simple cuboidal

Q444. Histological examination of the ovary shows a large antral follicle. Which cells directly produce estrogen?

A) Oocytes

✓ B) Granulosa cells

C) Fibroblasts

D) Macrophages

Q445. A biopsy from the uterus during the proliferative phase shows regeneration of endometrium. Which hormone predominates?

A) Progesterone

✓ B) Estrogen

C) Testosterone

D) Oxytocin

Q446. Vaginal epithelium is classified as:

A) Simple columnar

B) Transitional

✓ C) Stratified squamous non-keratinized

D) Pseudostratified ciliated

Q447. A patient with chronic bronchitis develops excessive mucus secretion. Which cells increase in number?

A) Basal cells

B) Clara cells

✓ C) Goblet cells

D) Type II pneumocytes

Q448. A patient with chronic vitamin B12 deficiency has degeneration of the spinal cord. Gastric biopsy reveals destruction of parietal cells. Which substance normally secreted by these cells is necessary for vitamin B12 absorption?

A) Pepsinogen

B) Gastrin

✓ C) Intrinsic factor

D) Mucin

Q449. Histological examination of a renal corpuscle demonstrates thickening of the glomerular basement membrane and fusion of podocyte foot processes. Which function is primarily impaired?

A) Tubular secretion

✓ B) Selective filtration of plasma

C) Reabsorption of glucose

D) Concentration of urine

Q450. A patient with chronic smoking history develops squamous metaplasia of respiratory epithelium. Which normal defense mechanism is most directly lost due to disappearance of cilia?

A) Surfactant production

✓ B) Mucociliary clearance

C) Antibody synthesis

D) Vasodilation